Preference for human or algorithmic forecasting advice does not predict if and how it is used

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Abstract

Past research has found that people treat advice differently depending on its source. In many cases, people seem to prefer human advice to algorithms, but in others there is a reversal and people seem to prefer algorithmic advice. Across two studies, we examine the persuasiveness of, and judges' preferences for, advice from different sources in forecasting geopolitical events. We find that judges report domain-specific preferences, preferring human advice in the domain of politics, and algorithmic advice in the domain of economics. In Study 2 participants report a preference for hybrid advice, that combines human and algorithmic sources, to either source regardless of domain. More importantly, we find that these preferences did not affect how persuasive advice from these different sources was, regardless of domain. Judges were primarily sensitive to quantitative features pertaining to the similarity between their initial beliefs and the advice they were offered, such as the distance between them and the relative advisor confidence, when deciding whether to revise their initial beliefs in light of advice, rather than the source that generated the advice.

Keywords: advice taking, algorithm aversion, algorithm appreciation, forecasting, hybrid forecasting, trust**.**

1 Introduction

When seeking assistance for difficult judgments, people can rely on many different sources. They can consult with experts they believe to be better informed than they are, larger groups of people to obtain a consensus, and increasingly even sophisticated statistical models which provide objective estimates based on specified criteria. This is particularly pertinent in the domain of forecasting economic and political events, such as those used in geopolitical forecasting tournaments (Himmelstein et al., 2021; Mellers et al., 2015; Morstatter et al., 2019; Tetlock & Gardner, 2016), which can be highly specialized and idiosyncratic. Individuals may feel comfortable relying on their own knowledge to forecast the outcome of a local election, but not when asked to predict election results in a country on the other side of the world. Investors may feel familiar with a particular class of commodities, but what if they are presented with a new opportunity in a field with which they lack familiarity? In situations such as these, it pays to use the best sources of advice available.

A growing body of evidence suggests that there are situations in which people mistrust the predictions of algorithms, even when the algorithms are known to be highly accurate, a phenomenon dubbed *algorithm aversion* (Burton et al., 2020; Dietvorst et al., 2015; Önkal et al., 2009). However, algorithm aversion is not universal or immutable. It is often exacerbated when people witness algorithms make errors (Dietvorst et al., 2015; Prahl & Van Swol, 2017), allowing human judges a measure of control of the algorithmic output can counteract it (Dietvorst et al., 2018), and a Logg et al. (2019) even identified several cases where people found algorithmic outputs more persuasive than human advice, a phenomenon labelled *algorithm appreciation.* Though there are large differences in the designs of these studies, they are often interpreted as yielding conflicting results (Hou & Jung, 2021), and several recent reviews attempt to synthesize these findings in detail (Burton et al., 2020; Mahmud et al., 2022).

This issue is of particular relevance to applications of *hybrid forecasting*, a topic recently explored in a large scale, longitudinal geopolitical forecasting contest known as the Hybrid Forecasting Competition (HFC) (Himmelstein et al., 2021; Morstatter et al., 2019; Zellner et al., 2021). The goal of the HFC was to combine human judgment and algorithmic forecasts to produce a superior alternative to either source of information on its own. One approach to this problem is to scaffold human judges with algorithmically generated forecasts, to aid in their judgments[1]. This work explores when, and how, human judges make use of advice from algorithmic and human sources in a geopolitical forecasting context. This includes (a) testing whether judges show aversion to, or appreciation of, algorithmic advice; (b) under what circumstance are judges more likely to accept different sources of advice and (c) what features of the advice itself make judges more likely to update their beliefs.

1.1 Trust and Persuasion

The most common elicitation method in forecasting studies is to ask judges to provide a probability distribution that represents their subjective beliefs about the likelihood of various outcomes (Morstatter et al., 2019; Tetlock et al., 2014). A straightforward method for incorporating algorithmic inputs in such cases is to provide forecasters with access to the predictions of forecasting algorithms and allow them to combine their prior beliefs and this new information. Such a process requires trusting the information and its source.

Trust can be defined as a willingness to allow one's own agency to be vulnerable or malleable to an external party or interest (Hardin, 1993; Mayer et al., 1995). In other words, trust

[1] We will henceforth use "algorithmic scaffolding" as an umbrella term for research involving using algorithms to aid human judgment, including research on both the phenomena of algorithm aversion and appreciation.

in a source is a necessary condition for judges to accept aid from it. We define the influence of new information on probabilistic judgments in this context as *persuasion*[2]. Empirical studies have shown that trust is necessary for persuasion to occur, but it is not sufficient, and that other factors also influence persuasion (Schul & Peri, 2015; Sniezek & van Swol, 2001; Van Swol & Sniezek, 2005).

Research on the persuasiveness of algorithms involves eliciting individual judgments, offering the judges advice, and studying if, how, and when, they update their original beliefs. This paradigm is known as a Judge Advisor System (JAS) (Logg et al., 2019; Sniezek & Buckley, 1995; Sniezek & van Swol, 2001). Logg et al. (2019) studied whether judges were more influenced by algorithmic or human advice across a number of domains. This work included an experiment which featured forecasts about two geopolitical events and one economic event (Experiment 4). Other experiments conducted by Logg et al. (2019) focused on a range of judgmental tasks, including estimating the weight of a person based on a photograph, predicting song rankings, and predicting attractiveness ratings. Across the different tasks, judges were more influenced by algorithmic than human advice, including when making geopolitical forecasts (though, interestingly, this result did not appear to generalize to domain experts).

Much of the literature on attitudes to algorithms involves tasks that do not necessarily focus on persuasiveness. This includes studies where judges are asked to express their willingness to rely on an algorithmic prediction over their own judgment (Dietvorst et al., 2015, 2018), and tasks in which judges rate how accurate they believe algorithms will be, compared to other human judges (Castelo et al., 2019). In addition, these studies often provide different pieces of information about performance to the judges. For example, Dietvorst et al. (2015) allowed the

[2] This view is related, but not identical, to the social psychological literature on persuasion, which is broader and encompasses many forms of attitude change in light of social information, see Wood (2000) for a review.

judges to see algorithms err, and Dietvorst et al. (2018) allowed judges various levels of autonomy in modifying algorithmic predictions. The behaviors observed in these tasks reflect the judges' trust in algorithms, but do not allow researchers to directly measure the persuasiveness of algorithmically generated information.

It is well established that verbal statements of preferences or beliefs are not always congruent with observed behavior (Diekmann & Preisendörfer, 1998; Ericsson & Simon, 1980; Nisbett & Wilson, 1977), and that explication of preference can even modify some behaviors (Wilson & Schooler, 1991). Indeed, this behavior-belief (in)congruity is another dimension of algorithmic advice taking that is under-studied, and we believe that this may lead to false generalizations about the persuasive impact of algorithms. Past studies have found that statements or presumptions of trust in advisors do not necessarily correspond how influential their advice is in belief revision (Goodwin et al., 2013; Önkal et al., 2017). More specifically, higher levels of trust did not meaningfully increase persuasiveness compared to controls, though lower levels of trust did, albeit slightly. This is consistent with the notion that a degree of trust is necessary for persuasion, but trust beyond a certain baseline may have less impact. However these studies focused on human, rather than algorithmic advisors. In this paper, we measure persuasiveness of advice from different sources, and elicit ratings of how accurate people believe those sources are likely to be, to help disentangle the role of trust in the persuasiveness of algorithmically generated information.

1.2 Forecasting Domains

Most studies on algorithmic scaffolding have focused on questions involving personal outcomes of individuals, or predictions about people's aggregate preferences. For example, Dietvorst et al. (2015) asked participants to forecast the academic performance of MBA application candidates, Logg, et al. (2019) asked participants to forecast song rankings and romantic attractiveness ratings, and Kaufmann and Budescu (2020) asked teachers to predict students' performance. Castelo et al. (2019) showed that algorithm aversion was greater in tasks that were perceived as more subjective (e.g. recommending music, predicting how funny a joke would be rated) and weaker in tasks rated as more objective (e.g. predicting the stock market, diagnosing a disease), even when the algorithm was demonstrated to outperform human advisors.

Forecasting geopolitical and economic events is a slightly different setup. The questions are highly impersonal, and they have ground truths that are eventually revealed and allow objective quantification of the forecasts' accuracy. The distinction between subjective and objective items (Castelo et al. 2019) has a parallel in forecasting. There is a rich literature on the use of algorithms to predict economic outcomes, often based on historical data and time series models (Franses et al., 2014; Huang et al., 2005; Pai & Lin, 2005; Todd, 1990; Zellner et al., 2021). Computational modeling of geopolitical events is a more difficult task, with the possible exception of national elections (Morstatter et al., 2019). Whereas economic outcomes are often driven by stationary quantitative processes (e.g. exchange rates between currencies), political questions tend to focus on discrete, one-off events (e.g., will a governing body pass a proposed legislation during the current term?). This distinction leads to a prediction that judges would be more likely to prefer and use algorithmic advice in economic forecasting for processes that are historically stable and well understood, as opposed to forecasting of political events which no not have obvious historical base rates.

1.3 Time Horizons and Feedback

The personalized nature of the events that in which algorithm aversion is often found come with an additional constraint: the topics typically forecasted are unrelated to, or invariant across different time horizons. Judging who is a good MBA candidate or identifying an attractive individual should not change substantially over time. However, when forecasting commodity prices, military incursions, or rates of pandemic infections, the amount of time remaining before the ground truth resolution plays a critical role in judges' beliefs and confidence in those beliefs.

In past geopolitical forecasting research, a commensurate increase in both confidence and accuracy has been observed in forecasts based on shorter rather than longer time horizons (Chen et al., 2016; Himmelstein et al., In Press; Moore et al., 2016). This is reflected in people's confidence in their own beliefs, but not necessarily their willingness to accept advice. This raises an interesting question: will judges be more willing to accept advice, regardless of source, for longer time horizons when their confidence in their own beliefs is weaker? Moreover, will this willingness vary depending on whether the advice comes from an algorithm or human?

According to construal level theory (Trope & Liberman, 2003, 2010) temporally, and thus psychologically distant objects are conceptualized using greater abstraction and generality (high construal), and more proximal objects are conceptualized using greater incidental detail (low construal). Another way to frame this question is to ask whether people will be more willing to accept algorithmic advice in cases where forecasts are perceived as less sensitive to immediate detail, and more contingent on general principles. The underlying logic of this prediction is that we tend to think of algorithms as unaffected by transient and local incidental details, which are more likely to be relevant in the short term, but less likely to be relevant in the long term.

Much past research on algorithm aversion has involved scenarios in which judges received immediate feedback and studied how behavior changes after seeing the (in)effectiveness of different types of advisors. People did not typically express algorithm aversion until they encountered cases where the advice was poor and the algorithms erred (Dietvorst et al., 2015; Prahl & Van Swol, 2017). Conversely, observing successful algorithmic advice reduced algorithm aversion (Filiz et al., 2021). For example, when asking judges to predict the results of MBA applications, Dietvorst et al. (2015) were using cases where outcomes were known, and judges learned that they would receive immediate feedback during the task. Geopolitical forecasting scenarios involve events with specific, and often long time horizons for which meaningful feedback is not immediately available. Professional forecasters (in intelligence, weather, finance, sports and other domains) are rarely asked to predict outcomes which are already known. When faced with questions about uncertain future events, knowing immediate feedback is not available, will judges treat algorithmic advice differently than advice from human experts?

1.4 Advice Description

A final issue that may affect judges' willingness to accept advice from different sources involves descriptor terminology. Algorithmic advice is often described using terms such as "a statistical model" (Dietvorst et al., 2015; Kaufmann & Budescu, 2020; Logg et al., 2019; Önkal et al., 2009) or, simply, "an algorithm" (Logg et al., 2019), and human advice is described using terms such as "a financial expert" (Önkal et al., 2009), "another participant" (Logg et al., 2019) or, in some cases, as an aggregate of human judgments, namely "the average estimate of other forecasters" (Logg et al., 2019). Recently, buzzwords such as "artificial intelligence (AI)", "deep learning" and "neural network" have seen increasing colloquial use. These terms have well-defined technical meanings, but it is unknown whether they sound more trustworthy to a layperson than a simpler descriptor such as "a statistical model". Thus, in addition to studying the effects of advice source, content domain, and time horizon, Study 1 also included a "buzzword" condition to test whether this source is treated differently.

1.5 Benefits of Advice

In addition to studying how people use advice, we test whether, and how much, advice helps judges. In principle, the more accurate advice is, the more accurate judges using advice should become. Past research suggests that people do indeed make use of advice and that doing so benefits accuracy (Bonaccio & Dalal, 2006; Soll & Larrick, 2009; Soll & Mannes, 2011; Yaniv, 2004; Yaniv & Milyavsky, 2007). However, advice utilization tends to be suboptimal (Soll & Larrick, 2009; Yaniv, 2004), and people tend to weight advice more heavily for tasks perceived as being difficult (Gino & Moore, 2007). Normatively, one would expect judges to make similar use of algorithmic advice as they do with equally accurate human advisors. However, a consequence of algorithm aversion is that algorithmic advice may be discounted more heavily than human advice, leading to reduced benefits. Likewise, a consequence of algorithm appreciation may be increased benefits to accuracy to algorithmic advice (Logg et al., 2019).

2 Study 1: Algorithmic Scaffolding in Various Domains and Time Horizons

Study 1 sought to test whether, and how, judges use advice generated by algorithmic or human sources in revising their own beliefs under several conditions. These conditions varied in terms of the domain of the events being forecasted, the time horizon to forecast resolution, and the way the source of advice was described. In a separate experimental task, we measured the judges' self-reported preferences for advice from different sources across these different conditions, without eliciting actual predictions about the events from the judges.

2.1 Methods

2.1.1 Generating Advice

Prior to launching the study, we conducted a pilot study to generate the advice shown to judges. The pilot was conducted during the second week of August in 2018. Five volunteers (three men, two women, mean age 35.2, SD = 9.5) and sixty recruits from Amazon Mechanical Turk (20 women, 40 men, ages 22 to 57, mean age = 35.0, SD = 8.8, paid $3 for participation) were tasked with making forecasts on 25 different questions inspired by those administered during HFC: 10 questions about economics, 10 about geopolitics, and five labeled “other”[3]. See Supplementary Materials for a list of all items. To incentivize accuracy, we informed the judges that the three most accurate would receive a $25 bonus.

Each question had either two or four mutually exclusive and exhaustive response categories (bins), to which participants assigned probabilities (expressed as percentages) totaling 100%. We selected questions where we could manipulate the time horizon, so they could be plausibly asked with different resolution times (e.g., questions about the value of the stock market, the number of earthquakes or terrorist attacks recorded globally). Participants were assigned to one of two time-resolution sequences. In one sequence, all odd numbered items were assigned a resolution date during September 2018 (the “short” time horizon) and even numbered items were assigned a resolution date during November 2018 (the “long” time horizon). In the alternate sequence the ordering was reversed, to ensure a similar number of forecasts were generated for each question and time horizon combination. See Figure 1 for an example of a forecasting question from the pilot.

For each question, the advisory forecasts were simply the mean values assigned to each response option by the 65 judges in the pilot study. This ensured that the forecasts would be

[3] Questions in the “other” category included items about public health/disease, earthquake frequency, and technology/data breaches

plausible and, based on the theory of the wisdom of crowds (Atanasov et al., 2017; Budescu & Chen, 2014; Surowiecki, 2005), they would be likely to be accurate as well.

2.1.2 Participants

Participants in Study 1 were 250 recruits from Amazon Mechanical Turk (mean age = 37.4, SD = 11.8, 42% women). They were paid $4 for their participation and were informed that six additional $25 bonuses would be awarded to the most accurate judges; three for the most accurate judges based on their initial forecasts, and three for the most accurate updated forecasts after viewing advice. This was done to incentivize judges to produce accurate forecasts both prior to and after seeing advice.

2.1.3 Procedure

We ran two experimental tasks: a persuasive *JAS task* and a self-reported *preferential task*. The JAS task consisted of a 2 x 2 x 3 x 6 design where we manipulated the advice source (algorithm vs. human, within subject), time horizon (short vs. long, within subject), question domain (economics, politics, other within subject), and description of advice source (between subjects). Each judge viewed 20 forecasting questions taken from the pilot study (eight economics, eight politics, four other[4]; see Supplementary Materials), and answered 25% of all questions for each combination of advice source and time horizon[5]. Each judge was assigned a descriptor term for human and algorithmic advice that was consistent throughout. There were two human advice descriptors: single (an “expert in the field”) and ensemble (“an average of several experts in the field”). There were three algorithmic advice descriptors: single (“a statistical model”), ensemble (“an average of several statistical models”), and buzzword (“a deep

[4] One question in the “other” category had a programming error was removed from the analyses

[5] Time horizon and advice source were inadvertently linked in programming the experiment, and as a result this factorial combination was between-subjects

neural network”). The six description conditions consisted of all unique combinations of the two human and three algorithmic descriptors. The order of the 20 items was randomized for each subject.

For each question, judges made an initial forecast, as in the pilot study (see Figure 1). After submitting the forecast, they were shown the advisory forecast, and given the opportunity to revise their initial forecast (which was also displayed back to them) based on this new information (see Figure 2). The advisory forecast was described (matching their description condition) as either from a human or algorithmic source. The actual advice forecast was the mean forecast from the pilot for that question and time horizon, for all descriptions. Technically, describing the advisory forecasts as generated from a “statistical model” or “expert in the field” could be considered accurate, but calling it a “deep neural network” could not. Thus, at the end of the study all judges were debriefed on how the advisory forecasts were generated.

The secondary, preferential, task used 10 different questions (Four economics, four geopolitics, two other; see Supplementary Materials). Judges used a slider to indicate the degree to which they thought a human or algorithmically generated forecast would be better suited to accurately forecast each event. Each judge saw half of all questions for each time horizon, and the description of human and algorithmic advice matched their description condition assignment from the behavioral task. The order of the 10 items was randomized for each subject. Figure 3 presents an example of the task.

2.1.4 Belief Revision Measurement

Past studies on persuasiveness of algorithmic scaffolding typically focused on point estimates of a continuous variable (Logg et al., 2019; Prahl & Van Swol, 2017), rather than probability distributions across a series of discrete outcomes. For example, one of the studies reported by Logg et al. (2019) asked participants to estimate attractiveness ratings, on a 100-point scale. Post-advice belief revision is often measured by using variants of a metric known as *weight of advice* (WOA) (Bonaccio & Dalal, 2006; Logg et al., 2019; Prahl & Van Swol, 2017; Soll et al., 2021). WOA is defined as the ratio of differences (signed distances) between a judge's initial estimate (F_1), a judge's revised estimate (F_2), and the advisor's recommended estimate (F_a). If the advice, (F_a), is different from one's original estimate, (F_1), then:

$$WOA = \frac{F_2 - F_1}{F_a - F_1}$$

If the final forecast (F_2), is bracketed by the initial forecast (F_1), and the advisor forecast (F_a), it can be represented as a weighted combination of the two, and WOA takes on a value between 0 (rejection of the advice) and 1 (adoption of the advice). If WOA is negative, it suggests a judge was *dissuaded* by advice, as their final forecast is *farther* from that value than their initial forecast; and if WOA is greater than 1, it suggests over-discounting of one's initial belief. In some studies, the absolute values of the numerator and denominator (two distances) are used, meaning WOA (denoted below aWOA) cannot be negative, and represents the magnitude by which a judge revises their belief following advice, regardless of the direction of the change.

$$aWOA = \frac{|F_2 - F_1|}{|F_a - F_1|}$$

Normatively, unless a judge believes advice is intentionally deceptive, or did not report their true beliefs at F_1, a bracketed F_2 strictly dominates an unbracketed one with regard to expected accuracy (Larrick & Soll, 2006). As such, in many JAS studies, observations where F_2 is not bracketed by F_1 and F_a are Winsorized to the nearest end point (0 or 1), such that all WOA (and aWOA) values lie in the (0, 1) interval (e.g. Logg et al., 2019; Schultz et al., 2015). Bonaccio and Dalal (2006) provide various alternative recommendations as well, including analyzing JAS data twice: with and without the unbracketed observations, to determine how influential they are.

In our study, the judges are estimating probability distributions across a partition of either *C* = 2 (binary items) or 4 (quaternary items) events. In the binary case, a version of WOA can be easily derived, as P(Event 1) = 1 – P(Event 2). However, in the quaternary case, the standard WOA measure does not apply, as the full probability distribution cannot captured by a single value. Instead, we rely on the Euclidean distance[6] between the two discrete probability distributions.

$$D(p,q) = \sqrt{\sum_{i=1}^{c}(p_i - q_i)^2}$$

We generated a distance-based metric similar to WOA

$$DWOA = \frac{\mathrm{D}(F_1, F_2)}{\mathrm{D}(F_1, F_a)}$$

DWOA is non-negative, and when *C* = 2, DWOA = aWOA. However, when *C* > 2, it is impossible to meaningfully measure a “direction” and distinguish between positive and negative values. As such, for many analyses we follow the recommendation of Bonaccio and Dalal (2006) and exclude cases where a judge’s revised belief is more distant from either the advice or from their initial belief than their initial belief was from advice[7]. Figure 4 represents this choice graphically in the *C* = 3 case.

[6] We also considered two other distance options, absolute distance and Hellinger distance, each of which was highly correlated with Euclidean distance (see Supplementary Materials).
[7] Statistical results which include these observations are included in Appendix C with the supplementary materials. They do not alter any key conclusions.

2.1.5 Measuring Accuracy

Accuracy of probabilistic forecasts is often assessed by Brier Scores (BS) (Brier, 1950). The BS is a strictly proper scoring rule applicable to discrete probability distributions (Gneiting & Raftery, 2007; Merkle & Steyvers, 2013). The BS are squared Euclidean distances, where *p* is a probability distribution based on the judges' estimates, and q is a singleton where one bin (the eventual "ground truth") takes a value of 1 and all the other bins are 0.

The traditional Brier score applies to questions with any number, *C (C* =2,3…), of unordered outcomes. However, consider the example in Figure 1, where the options have a natural order. If we assume that the truth winds up being "Greater than 220", it makes sense to consider a forecast that treats "Greater than 200, but less than 220" as the most likely outcome superior to one that treats "Less than or equal to 180" as the most likely outcome. Traditional Brier scores are agnostic to this distance to ground truth-based distinction, but a variant of the Brier score that respects this ordinality can be used (Himmelstein et al., 2021; Jose et al., 2009).

2.2 Results

2.2.1 Acceptance of Advice

Overall, across (250 judges x 19 items =) 4,750 forecasts, F_2 was different from F_1 49.6% of the time, indicating the about half of the forecasts were updated after viewing advice. This effect was consistent across experimental conditions, ranging between 43% and 59% across all individual cells, and between 46% and 53% across all marginal conditions individually.

In the N = 2,357 cases in which F_2 was different from F_1[8], $\mathrm{D}(F_2, F_a) \leq \mathrm{D}(F_1, F_a)$ 2,300 (97.6%) times, meaning that F_2 was at least as close to F_a as F_1 to F_a. This effect was consistent

[8] In the remaining 2,393 cases F_1 = F_2. In 18 cases, DWOA was undefined, as F_1 = F_a as well (all these cases were binary items).

across experimental conditions, ranging between 96% and 99% across cells, and between 97% and 98% across marginal conditions individually. Similarly, $\mathrm{D}(F_1, F_2) \leq \mathrm{D}(F_1, F_a)$ 1,957 (83.0%) times, meaning that F_2 was at least as close to F_1 as to F_a to F_1. This effect was consistent across experimental conditions, ranging between 78% and 88% across cells, and 81% and 85% across marginal conditions. There were also 31 cases (1%) where $\mathrm{D}(F_2, F_a) > \mathrm{D}(F_1, F_a)$ and $\mathrm{D}(F_1, F_2) > \mathrm{D}(F_1, F_a)$, meaning F_2 was further from both F_1 and F_a than they were from each other. This means that in total, there were 424 cases (18% of N = 2,357 revisions, 9% of N = 4,750 total observations) where $\mathrm{D}(F_2, F_a) > \mathrm{D}(F_1, F_a)$ or $\mathrm{D}(F_1, F_2) > \mathrm{D}(F_1, F_a)$. These cases were removed from subsequent analyses involving DWOA (analyses with these observations included can be found in Supplementary Materials).

To determine if people were more likely to discount their prior beliefs than the advice, for each judge, we totaled the number of times $\mathrm{D}(F_1, F_2) > \mathrm{D}(F_1, F_a)$ as well as the number of times $\mathrm{D}(F_2, F_a) > \mathrm{D}(F_1, F_a)$ and performed a sign test. The median number of times people discounted their prior beliefs (1) was significantly larger than the median number of times they discounted the advice (0), S = 4, $p < .001$.

Figure 5 displays mean $\mathrm{D}(F_1, F_a)$ and $\mathrm{D}(F_2, F_a)$ for each item and time horizon (across both sources of advice), and paints a clear picture: $\mathrm{D}(F_2, F_a)$ was consistently smaller than $\mathrm{D}(F_1, F_a)$, reflecting judges' willingness to take advantage of the advice they were offered. The mean effect of advice was remarkably stable across items and time horizons.

2.2.2 Buzzword Effect

We tested the effect of the buzzword advice description condition. This term was only applied to algorithmic advice, so we analyzed only the 2,152 relevant cases. We fit a one-way mixed effects ANOVA on DWOA with random intercepts for participants and items. The effect of advice description was not significant, $F(2, 247.95)$[9] = 1.36, p = .26. We conclude that the "buzzword" condition was not different from the other machine conditions.

2.2.3 Ensemble Effect

We also tested the effect of the "ensemble" descriptor, which was used both for the human and algorithmic advice, but not for the buzzword advice, so we analyzed only the 3,597 relevant cases. We fit a two-way mixed-effects ANOVA with random intercepts for participants and items, with advice description (single vs. ensemble), advice source (human vs. algorithm), and their interaction predicting DWOA. The interaction between description and source was not significant, $F(1, 3{,}325.8) = 0.53$, $p = .46$. The main effect of description was not significant, $F(1, 3{,}410.9) = 1.47$, $p = .23$ and, most importantly, the main effect of source was not significant, $F(1, 3{,}472.8) = 0.27$, $p = .60$. Thus, we found no ensemble effect.

Because neither the buzzword nor the ensemble manipulations significantly affect DWOA, the advice description effects are omitted from the remainder of this analysis, and we focus on advice source, time horizon, and item domain.

2.2.4 Advice Source, Time Horizon, and Domain

We fit a mixed effects ANCOVA on DWOA using advice source, time horizon, and domain and with random intercepts for items and participants. We also included demographic and advice covariates. Demographic covariates included participants' age, gender, and whether they had taken a class in statistics[10]. We also included an interaction between age and advice source, to test whether amenability to algorithmic advice varied by age. Advice covariates included the (Kendall) rank order correlation between the C bins for F_a and F_1, and the relative

[9] All degrees of freedom adjustments use the Satterthwaite method

[10] Analyses without covariates are reported in the Supplementary Materials. None of the key statistical results meaningfully change.

peakedness of F_a and F_1. The rank correlation between the two forecasts can be seen as a measure of agreement in the discrimination of the likelihood of each option (we refer to this variable as *discrimination agreement*). Peakedness is the difference between the probability assigned to the most likely option for each item and the average of the other (*C – 1)* probabilities. It is a simple measure of the certainty a judge places of the most likely outcome:

$$Peakedness = Max(P) - \frac{[1 - Max(P)]}{C - 1} = \frac{CMax(P) - 1}{C - 1}$$

Relative peakedness is the natural logarithm of the ratio of the Peakdness of F_1 and F_a. Positive values represent cases where the judge's forecast is more peaked than the advice, and negative values represent cases where the advice was more peaked than the judge.

$$Relative\ Peakedness = \log\left(\frac{Peakedness_{T1}}{Peakedness_a}\right)$$

None of the factors manipulated experimentally – advice source, time horizon, item domain – nor any of their interactions were significant: Three-way interaction between advice source, time horizon, and domain (F(2, 4,035.3) = 0.23, p = .79); two-way interaction between domain and time horizon (F(2, 4,055.5) = 0.16, p = .85); two-way interaction between advice source and domain (F(2, 4,038.5) = 0.01, p = .99); two-way interaction between advice source and time horizon (F(1, 252.9) = 2.33 , p = .13); effect of domain (F(2, 16.9) = 0.21, p = .81); effect of time horizon (F(1, 4,060.8) = 0.88, p = .35); and effect of advice source (F(1, 4,040.5) = 1.61, p = .20).

The only covariate that was significant was the discrimination agreement (β = -0.17, F(1, 3,783.7) = 479.04, p < .001, Standardized β = -0.27, partial ω^2 = 0.11). The judges revised their original forecasts more when the advice disagreed with their rank ordering of the options in terms of their likelihood. This indicates a potential *counterfactual effect* of advice. When the advice information contradicts the judges' expectation in terms of the most and least likely events, it is interpreted as a stronger signal than when the information indicates their probabilities are miscalibrated, but the rank order is similar.

2.2.5 Preferential Task

Next, we tested the effect of the various manipulations on the preference reports (see Figure 3). The ratings were values between 0 and 100, where lower values indicate preference for human advice and higher values preference for machine advice (these values were not displayed to participants). The mean rating across all items was 55.06 (SD = 30.49), indicating a slight preference for algorithmic advice, but the fixed intercept in an unconditional model with random intercepts for participants and items was not significantly different from 50, $t(9.37) = 0.97$, $p = .36$.

We fit a two-way mixed ANOVA on the preference ratings with item domain and time horizon serving as predictors, with random intercepts for items and participants. We also included covariates for how the two sources of advice were described (single, ensemble, neural network), as well as age, gender, and whether the judge had taken any statistics classes. The interaction between item domain and time horizon was not significant, $F(2, 2{,}237.1) = 0.02$, $p = .98$. The main effect of time horizon was also not significant, $F(1, 2237.1) = 0.05$, $p = .88$. The main effect of domain was significant, $F(2, 7.0) = 24.39$, $p < .001$, partial $\omega^2 = .82$.

Follow-up pairwise comparisons for domain (with Tukey HSD adjustments) indicated the difference between economics and politics was significant, adjusted mean difference = 32.1, $t(7) = 6.95$, $p < .001$, 95% CI = [18.5, 47.5]; the difference between other and politics was significant, adjusted mean difference = -19.8, $t(7) = 3.51$, $p = .02$, 95% CI = [3.2, 36.5]; but the difference between economics and other was not significant, adjusted mean difference = 12.3,

$t(7) = 2.17$, $p = .14$, 95% CI = [-4.4, 29.9]. Confirming the pairwise comparisons, the Figure 6a shows that the significant effect of domain was driven by a preference for algorithmic advice for economic questions and human advice for politics questions.

The main effect of algorithm condition was also significant, $F(2, 243.1) = 9.52$, $p < .001$, partial $\omega^2 = .07$; but the effect of human condition was not, $F(2, 242.4) = 0.02$, $p = .87$. Follow up pairwise tests (with Tukey HSD adjustments) on the effect of algorithm condition revealed a significant difference between single model and neural network, adjusted mean difference = 7.37, $t(243.1) = 3.61$, $p = .001$, 95% CI = [2.55, 12.18]; and between ensemble of models and neural network, adjusted mean difference = 8.39, $t(243.1) = 4.07$, $p < .001$, 95% CI = [3.53, 13.26]; but not ensemble and single, adjusted mean difference =1.03, $t(243.1) = 0.51$, $p = .87$, 95% CI = [-3.78, 5.84]. Figure 6b confirms this result, showing people preferred algorithms slightly less in the buzzword condition than the single or ensemble conditions.

2.2.6 Persuasion-Preference Correspondence

For each participant, we computed the mean DWOA values for cases in which they saw each source of advice in the JAS task, and mean preference scores in the preference task, overall and for each domain. Figure 7 shows the joint distributions and Kendall's rank correlations in each case with bootstrapped 95% confidence intervals. If there were a persuasion-preference correspondence, we would expect this correlation to be positive in cases where participants saw algorithmic advice, and negative in cases in which they saw human advice, since higher preference ratings mean stronger preference for algorithmic advice. We did not find significant correlations between expressed preference for advice source and DWOA.

2.2.7 Benefits of Advice

Figure 8 shows the average Brier scores for each item and time horizon for (a) judges who did not update based on advice, (b) judges who did update, pre and post advice, as well as (c) the Brier score of the advice forecasts[11]. The results confirm that the advice received was relatively accurate, and that, on average, advice benefited judges who used it. Interestingly, the judges who did not update tended to be more accurate at time 1 (T1) than those who did, suggesting they recognized the cases when they needed advice. This is further quantified in Table 1, which shows the average Brier scores in each case for various experimental conditions. While judges who updated tended to be more accurate at time 2 (T2) than those who did not, the difference is much smaller than the gap between T1 and T2 for those who did update.

2.2.8 When Do Judges Update?

Recall that judges updated their forecasts after viewing advice approximately half the time. This indicates a high variance in the participants' decision whether to revise following advice. This begs the question whether the initial decision to update represents a separate process from how much they update. An equally important question is whether one can predict which judges will revise their estimates, and under what circumstances?

Research on similarity between preliminary beliefs and advice suggests judges *egocentrically trim,* i.e., they discount advice more heavily based on how discrepant it is from their own beliefs (Fink et al., 1983; Yaniv & Milyavsky, 2007). Other accounts suggest the effect is curvilinear, and that people may not modify their beliefs on the basis of extremely proximal advice either (Allahverdyan & Galstyan, 2014; Schultze et al., 2015; Whittaker, 1963). The JAS

[11] This analysis does not involve DWOA, so it includes observations omitted from DWOA analyses

literature also indicates that acceptance of advice is driven, to a large degree, by advisor confidence (Sniezek & van Swol, 2001; Van Swol & Sniezek, 2005).

Our accuracy analysis indicates that judges more frequently updated forecasts that were relatively inaccurate at T1. This could reflect a self-awareness of their knowledge about an item (or lack thereof), and recognition of the greater potential benefits of advice in some cases more than others. We did not elicit confidence ratings for each forecast, but the peakedness measure serves as a proxy of a judge's relative confidence in the likelihood of the various outcomes, as compared to the information present in the advisory forecast.

To test these new exploratory hypotheses, we fit mixed effects logistic regressions seeking to predict whether judges updated their forecasts after viewing advice. We fit models with random intercepts for judges and items on the N = 4,732 cases where $F_1 \neq F_a$[12]. We distinguish between three classes of predictors (a) personal demographic information (age, gender, whether judges had taken a course in statistics), that is available prior to making a forecast; (b) the factors manipulated (advice source, item domain, time horizon), which reflect the information that was available to the judges when making the initial estimates and before having access to the advice; (c) variables that reflect the relationship between the initial estimates and the advice (relative peakedness, discrimination agreement, and $\mathrm{D}(F_1, F_a)$), which become available only after judges access advice. We also included the three-way interaction between the manipulated factors (b), a quadratic term for distance $\mathrm{D}(F_1, F_a)^2$ representing a possible non-linear effect of advice proximity, and two-way interactions between the belief-advice similarity variables (c).

[12] Once again, as this analysis does not involve DWOA, we do not discard any other observations

These three classes of predictors come into play at various points in time, so a comparison of the three models can pinpoint the time at which judges make the decision to consider advice. Likelihood ratio tests for the variables of interest are reported in Table 2 across the three models: Model 1 uses only individual differences; Model 2 adds information about the manipulated factors and Model 3 includes all the variables. No effects were statistically significant in Model 1 or Model 2, though domain and time horizon were significant in Model 3 once belief-advice similarity features were controlled for. The BIC index was notably lower, and AUC clearly higher in Model 3 than Models 1 or 2, suggesting that the decision to revise one's forecast occurs upon accessing the information contained by the advice, rather than as a result of demographic factors or the source of advice.

In Model 3, advice source and the various interactions between advice source, time horizon, and item domain were not significant. The main effects time horizon and item domain were significant, but the effects were quite small. Participants were slightly more likely to update for longer time horizons (51.1%) than for shorter ones (48.6%), and they were slightly more likely to update for questions about Economics (52.8%) compared with Politics (46.0%).

The main effects of relative peakedness, discrimination agreement, distance ($\mathrm{D}(F_1, F_a)$), its quadratic effect, and all their interactions were statistically significant except for the interaction between relative peakedness and discrimination agreement. Figure 9a displays the interaction between relative peakedness and $\mathrm{D}(F_1, F_a)$. When F_1 was more peaked than F_a, the effect of distance was weaker and less curved, such that even for moderate distances there was some uncertainty in whether participants would update. However, when F_1 was less peaked than F_a, the effect of distance was stronger and more curved, such that updates were highly likely in particular for moderate distances. Figure 9b shows the interaction between $\mathrm{D}(F_1, F_a)$ and discrimination agreement. The quadratic effect of distance is again present in this interaction; however, judges were much more likely to update when there was high disagreement, regardless of distance. Figure 9c shows the joint effects of disagreement and peakedness. Belief revision decreased in cases where the judge was more certain than the advice and when there was larger rank order agreement across the options. This suggests that either high advisor confidence or counterfactual advisor information are sufficient to encourage judges to revise their beliefs.

2.3 Discussion

The original goal of Study 1 was to compare the influence of information generated by algorithms to information generated by human experts, across different domains and time horizons in the context of forecasting geopolitical events that require item-specific research. Our design kept the advice fixed but manipulated its description. Remarkably, we observed no evidence that the description of the source of information, time horizon, or domain were associated with the advice persuasiveness, as inferred from the judges' willingness to revise their initial forecasts and the magnitude of belief revisions.

However, when simply asking judges to rate the expected accuracy of the different sources of advice we observed a very clear domain specific preference pattern—algorithm appreciation for economics and aversion for politics—which is consistent with the documented preference for algorithms in more objective domains (Castelo et al., 2019). One of the most interesting results of our work is the lack of correspondence between the behavioral and preference patterns across the two experimental tasks. The mismatch between behavior and self-reports of intentions in not new (Diekmann & Preisendörfer, 1998; Ericsson & Simon, 1980; Nisbett & Wilson, 1977), but this is the first documentation of this well-known cognitive incongruence in the context of forecasting algorithmic advice utilization, with clear implications for the hot “human-model competition”.

We have also shown that, on average, judges do a good job of identifying the cases where they should accept, and follow, advice in either domain. These results indicate that, when deciding how to revise their own beliefs, judges are more concerned with content, perceived quality, and relationship between their prior beliefs and the advice, than whether it came from a human or algorithm, even if they express greater trust in one source or the other for a particular topic.

We also manipulated the forecasts’ time horizons, but this factor did not appear to play a meaningful role in judges’ decisions in either task, and did not interact with the advice source. It is possible that the time horizon differential was simply not large enough. The short time horizon involved forecasts for events approximately two weeks in the future, while the long time horizon condition the events were approximately 10 weeks in the future. While these horizons were selected to reflect forecasting problems frequently found in forecasting tournaments, such as the HFC, other forecasting tasks, particularly involving economic analysis, can involve time horizons on the magnitude of years, rather than weeks or months (e.g. Himmelstein et al., In Press). Future studies should seek to determine whether a more prominent effect of time horizon would emerge if time horizon differences were of a different magnitude.

Two additional exploratory factors were manipulated: the effect of ensembling advisors (averaging several experts / statistical models) and the effect of using a buzzword (“deep neural network”) to describe algorithmic advice. Though we did not have a strong hypothesis about the former manipulation, we expected the buzzword terminology to increase trust in algorithms. Neither factor significantly predicted advice utilization, but there was a significant effect of algorithm condition in the preferential task: participants actually preferred the buzzword terminology slightly *less*. There are several possible explanations for this unexpected result. Perhaps participants associated the buzzword terminology with greater opacity, believing it to describe more of a “black box” algorithm, while the simpler terminology was more psychologically accessible and thus perceived as more transparent. Or perhaps the use of the buzzword elicited a sense of deception and reduced trust, as if it were like a bad marketing ploy. The size of the effect was relatively small, and these explanations are speculative, thus they warrant more dedicated investigation in future research.

3 Study 2: Replication with Hybrid Advice

In Study 1, we looked at the impact of advice from either a single human or algorithm, or an average of several humans or algorithms. Here we consider the impact of advice from an average of a human *and* an algorithm, which we refer to as *hybrid* advice (Zellner et al. 2021). We know that trust in the source of advice influences people’s willingness to accept it (Wang & Du, 2018), but that higher levels of trust do not always predict greater persuasion (Goodwin et al., 2013; Önkal et al., 2017). It is also unclear how judges will react to advice that combines various sources which may be perceived as having differing levels of trustworthiness. This question is particularly relevant in the context of the Hybrid Forecasting Competition, which studied the effects of aggregating human and algorithmic forecasts. Judges who are presented with hybrid advice may treat it as having the strengths of both without the weaknesses, or simply average their impression of the separate inputs individually. The next study will also allow us to test if key results of Study 1—the gap between trustworthiness ratings and persuasiveness, and alternative mechanisms driving belief revision involving similarities between prior beliefs and advice—replicate as well as generalize to the hybrid case.

3.1 Methods

Study 2 was conducted in early November of 2018 and used largely the same design as Study 1. In the interest of brevity, this section highlights the differences between the two studies.

3.1.1 Generating Advice

We used the same events as in Study 1, and the same advice. The main difference was that we dropped the time horizon manipulation. The advice forecasts for November resolutions, which were the long time horizon in Study 1, were set to resolve about two weeks from the launch of Study 2. As such, they became more similar to Study 1's short time horizon, and we did not include a new long time horizon.

3.1.2 Participants

We recruited 171 volunteers from Amazon Mechanical Turk (mean age = 36.6, SD = 10.3, 40% women). Participants were paid $4.50 for their participation, with the same accuracy incentives as in Study 1.

3.1.3 Procedure

We ran two experimental tasks: a behavioral *JAS task* and a self-reported *preferential task*. The original JAS task was simplified to a 3 x 3 design with manipulations for advice source (algorithm, human or hybrid, within subject) and question domain (economics, politics or other, within subject). Because the hybrid condition included multiple inputs by default, all three sources were described as ensembles (i.e., "an average of human experts / statistical models / human experts and statistical models"). Each judge viewed 15 of the forecasting questions used in the pilot study (six economics, six politics, three other; see Supplementary Materials). Five of the 15 questions were accompanied by advice from each of the different sources. The forecast elicitation was identical to Study 1 (see Figures 1 and 2). The order of the 15 items was randomized.

The preferential task used 10 additional forecasting questions (Four economics, four geopolitics, two other; see Supplementary Materials). Given that we had three advice sources, we used three separate sliders and asked the judges to rate how accurate they thought each source would be, using a 7-point scale (see Figure 10 for an example). The order of the 10 items was randomized.

After the forecasting stage, the participants were asked to complete several questionnaires measuring numerical reasoning ability, in addition to the demographic survey that was used in Study 1. This battery increased our information about individual differences, and it included an extended seven item version of the cognitive reflection test (CRT)[13] (Baron et al., 2015; Frederick, 2005), which measures people's ability to numerically reason in the face of intuitively appealing distractor option; and an eight-item subjective numeracy scale (SNS)[13] (Fagerlin et al., 2007), a self-report measure of numerical reasoning ability.

3.2 Results

3.2.1 Acceptance of Advice

Overall, across (171 judges X 15 items =) 2,565 forecasts, F_2 was different from F_1 48.9% of the time, replicating the result from Study 1 in which judges updated their forecasts about half the time. The results were again similar across conditions, ranging from 43% to 54% across cells and 46% to 51% across marginal conditions individually.

[13] Sample CRT Cronbach's $\alpha = .77$, SNS Cronbach's $\alpha = .87$

In the N = 1,255 cases in which F_2 was different from F_1[14], $\mathrm{D}(F_2, F_a) \leq \mathrm{D}(F_1, F_a)$ 1,224 (97.5%) times, meaning that F_2 was at least as close to F_a as F_1 to F_a. This effect was consistent across experimental conditions, ranging between 95% and 99% across cells, and between 97% and 98% across marginal conditions individually. Similarly, $\mathrm{D}(F_1, F_2) \leq \mathrm{D}(F_1, F_a)$ 1,034 (82.3%) times, meaning that F_2 was at least as close to F_1 as to F_a to F_1. This effect was consistent across experimental conditions, ranging between 77% and 90% across cells, and 80% and 85% across marginal conditions. There were also 12 cases (0.5%) where $\mathrm{D}(F_2, F_a) > \mathrm{D}(F_1, F_a)$ and $\mathrm{D}(F_1, F_2) > \mathrm{D}(F_1, F_a)$, meaning F_2 was further from both F_1 and F_a than they were from each other. This means that in total, there were 240 cases (19% of N = 1,255 revisions, 9% of N = 2,565) where $\mathrm{D}(F_2, F_a) > \mathrm{D}(F_1, F_a)$ or $\mathrm{D}(F_1, F_2) > \mathrm{D}(F_1, F_a)$. These cases were removed from analyses involving DWOA (analyses with these observations included can be found in Supplementary Materials). These results are highly consistent with Study 1.

To determine if people were more likely to discount their prior beliefs than the advice, we totaled the number of times for each person $\mathrm{D}(F_1, F_2) > \mathrm{D}(F_1, F_a)$ and $\mathrm{D}(F_2, F_a) > \mathrm{D}(F_1, F_a)$, and performed a sign test. The median number of times people discounted their prior beliefs (1) was significantly larger than the median number of times they discounted the advice (0), S = 8, p < .001. These results are also consistent with Study 1.

Figure 11 displays mean $\mathrm{D}(F_1, F_a)$ and $\mathrm{D}(F_2, F_a)$ for each item across all sources of advice. Replicating the picture from Study 1, we see that $\mathrm{D}(F_2, F_a)$ was consistently lower than $\mathrm{D}(F_1, F_a)$, reflecting people's willingness to take advantage of the advice. The mean effect of advice was remarkably stable across items in all domains.

[14] In the remaining 1,310 cases F_1 = F_2. In 8 cases, DWOA was undefined, as F_1 = F_a as well (all these cases were binary items).

3.2.2 Advice Source and Domain

We fit a mixed effects ANCOVA of DWOA using advice source and domain and with random intercepts for items and participants. We included the same covariates as in Study 1[15], namely advice covariates (relative peakedness and discrimination agreement) and demographic covariates (age, gender, and having taken a statistics class), as well as the CRT and SNS scores.

As in Study 1, neither the advice source nor the item domain nor their interaction were significant: two-way interaction between advice source and domain ($F(4, 2{,}102.8) = 0.39$, $p = .68$); effect of domain ($F(2, 13.0) = 0.29$, $p = .75$); and effect of advice source ($F(2, 2{,}101.6) = 0.21$, $p = .81$).

Replicating the pattern found in in Study 1, discrimination agreement was a significant covariate ($\beta = -0.16$, $F(2, 2{,}137.6) = 198.82$, $p < .001$, Standardized $\beta = -0.25$, partial $\omega^2 = 0.08$), confirming that judges were more willing to revise their original forecasts the less the advice agreed with them on the order of the most likely options, again indicating a potential *counterfactual effect* of advice. None of the other covariates were statistically significant.

3.2.3 Preferential Task

We tested the effect of the two manipulations on the expected accuracy for each source of advice on a scale of 1 to 7 (see Figure 10). Across all items, the mean ratings were 4.45 ($SD = 1.22$) for human advice, 4.84 ($SD = 1.23$) for algorithm advice and 5.19 ($SD = 1.18$) for hybrid advice, suggesting a preference for hybrid advice.

We fit a two-way mixed ANCOVA of these ratings with item domain and advice source, with random intercepts for items and participants, and CRT, SNS, age, gender, and stats experience as covariates. The two way interaction between item domain and advice source was

[15] Analyses without covariates are reported in the Supplementary Materials. None of the key statistical results meaningfully change.

significant $F(4, 4{,}943) = 72.54$, $p < .001$, partial $\omega^2 = 0.05$, as well as each marginal effect, advice source $F(2, 4{,}943) = 209.69$, $p < .001$, partial $\omega^2 = 0.08$; and domain $F(2, 7) = 17.80$, $p = .002$, partial $\omega^2 = 0.77$ None of the covariates were statistically significant. Simple main effects of advice source by domain with Tukey HSD adjustments indicated the only non-significant comparison was the difference between hybrid and algorithmic advice for economics (see Supplementary Materials for detailed statistical results of simple main effects analysis).

Figure 12 shows adjusted mean preference rating as a function of advice source and item domain. The preference for human advice for politics and algorithmic advice for economics was replicated. Importantly, across all domains, judges preferred hybrid advice over both human and algorithmic advice. Judges may believe hybrid advice captures the strengths of both sources of information, and/or potentially mitigates their weaknesses.

3.2.4 Persuasion-Preference Correspondence

To test the correspondence between preference for different advice sources and the degree to which judges used advice in the JAS task, we computed the mean preference ratings and DWOA values for each participant for each advice source, as well as for each advice source and domain. Figure 13 shows the scatterplots of the mean ratings and mean DWOAs, as well as their Kendall's rank correlations and 95% bootstrapped confidence intervals. Unlike in Study 1, where the ratings used a bipolar scale (human vs. algorithm), a positive correlation between DWOA and the various ratings would indicate correspondence in each case. None of the correlations were significant, indicating people's preferences for different advice sources were not meaningfully associated with their willingness to make use of those sources.

3.2.5 Benefits of Advice

Figure 14 shows the average Brier scores for (a) judges who did not update based on advice, (b) judges who did update, pre and post advice, and (c) the Brier score of the advice forecasts. Although the advice was again relatively accurate, in some cases, it had worse Brier scores than the mean accuracy of the individual forecasts. Recall that advice was generated prior to Study 1, while in Study 2, several weeks had passed since the pilot, giving the new judges an informational advantage (Himmelstein et al., In Press). However, the general patterns from Study 1 hold. Judges who did not update tended to be more accurate at T1 than those who did update, but when the advice was accurate, it benefited those who did update. This is illustrated in Table 3. Overall, updating benefited forecasters at a lower rate than in Study 1, but was still helpful on average. However, in Study 2, judges who did not update were, on average, more accurate than those who did, even at T2, suggesting they correctly diagnosed their advantage over the advice.

3.2.6 When Do Judges Update?

In Study 1, we found that several features of the relationship between the T1 forecasts and advice predicted well the judges' decision to update their beliefs at T2. We replicated this exploratory analysis in Study 2 using the N = 2,527 cases where $F_1 \neq F_a$ by fitting three models based on different classes of predictors: (a) personal demographic information (age, gender, and whether judges had taken a course in statistics; as well as SNS and CRT); (b) the factors manipulated (advice source, item domain, time horizon), which reflect the information that was available to the judges when making the initial estimates and before having access to the advice; (c) variables that reflect the relationship between the initial estimates and the advice (relative peakedness, discrimination agreement, and D(F_1, F_a)), which become available only after judges access advice. We also included the three-way interaction between the manipulated factors (b), a quadratic term for distance $\mathrm{D}(F_1, F_a)^2$ representing a possible non-linear effect of advice proximity, and two-way interactions between the belief-advice similarity variables (c). Likelihood ratio tests for the variables of interest, as well as BIC and AUC statistics for the three models are reported in Table 4. Once again, the (c) variables appear to be the key predictors, with effects largely replicating those of Study 1 (see Figure 15).

The one new predictor of interest was CRT score, which was significant in Model 3. The mean CRT score was 4.29 (SE = 0.06) in cases where judges updated their forecasts, compared to 3.96 (SE = 0.06) in cases where judges did not update. In Model 3, the coefficient for CRT score was 0.53, indicating the odds of making an update after viewing advice increased by 70% for each additional CRT item a judge answered correctly, controlling for the other variables in the model. This suggests that higher cognitive reflection predicts willingness to update. This is consistent with Mellers, Stone, Atanasov, et al., (2015), who found that belief updating mediates the relationship between cognitive reflection and accuracy.

3.2.7 Cross Validation of the Model

The modeling of the decision to update was an exploratory analysis initiated in Study 1. Given the strength of the effects, and their successful replication in Study 2, we decided to cross validate the logistic regression models across the two studies, to determine how well they predict out-of-sample data. We fit models based on both the Study 1 data (N = 4,732) and Study 2 (N = 2,527) data using all predictors used in both studies: advice source (omitting hybrid advice from the validation in Study 2, so the validation sample in this case was only N = 1,684), item domain, distance, distance squared, discrimination agreement, relative sharpness, age, gender, and having taken a stats class. The AUC for the responses from the Study 1 model on the Study 2 data was 0.62, slightly lower than the AUC from Model 3 in Table 4, but higher than Models 1 and 2 in

Table 4. The AUC of or the responses from the Study 2 model on the Study 1 data was 0.66, which matched the AUC for Model 3 in Table 2. This confirms that the fixed effects of these models are robust. There was a small amount of discrimination loss in the Study 1 model to Study 2 data, possibly due to the exclusion of the cases which included hybrid advice in Study 2.

3.3 Discussion

In Study 2, we introduced hybrid advice that was described as the average of humans and algorithms. While there is no evidence that judges were more, or less, willing to incorporate this type of advice into their forecasts, in the preferential task they systemically rated the hybrid advice as superior to either human- or algorithm-only advice across all domains.

Study 2 also confirmed that cognitive reflection predicts willingness to revise one's beliefs after viewing advice. Past research has found that cognitive reflection predicts belief revision in general (Mellers et al., 2015), but these results confirm this extends to belief revision based on new information.

With a few minor exceptions, nearly all of the key tests from Study 1 that were also conducted in Study 2 replicated successfully. This is especially important for the results that show similarities between initial judges' beliefs and the advice best predict their willingness to revise their beliefs, which were a loud but unanticipated result that emerged from the data collected during Study 1.

4 General Discussion

This research was motivated, in part, by the diverse findings regarding the role of algorithmic advice in the behavioral decision-making literature, which are often interpreted as conflicting (Hou & Jung, 2021). For example, people are reticent to trust algorithms over their own judgments even when it is normatively appropriate (Dietvorst et al., 2015), people rate the expected accuracy of humans and algorithms differently depending on domain (Castelo et al., 2019), and people often find predictions of algorithms relatively persuasive (Logg et al., 2019).

Our results suggest that these findings encompass several distinct, but not necessarily inconsistent phenomena. We find that people's ratings of the trustworthiness of human and algorithmic sources is not reflected in how persuasive those sources are. At first blush, this is a surprisingly result, but past research suggests differences in amount of trust above a certain level have limited impact on persuasion (Goodwin et al., 2013; Önkal et al., 2017), which is consistent with our findings. Despite differences in trust for different sources of information, trust remained sufficient for all sources for persuasion to occur at similar levels. Rabinovitch et al. (2022) recently found a similar pattern in their work about willingness to take advice in the context of difficult binary selection decisions. The participants in their experiments expressed a general preference for human advice, but were equally likely to accept and follow advice from either source. Our results also provide an alternative account of the mechanisms that drive persuasion. Judges' decision to revise their beliefs, as well as the magnitude of that revision, was driven by the information contained by the advice and its relationship to their prior beliefs, rather than its source.

4.1 Benefits of Advice

The benefits of advice were clear. The advice, which was based on an aggregate forecast from a pilot crowdsourcing experiment, was highly accurate. Consistent with past research on the benefits of using accurate advice (Bonaccio & Dalal, 2006; Soll & Larrick, 2009; Soll & Mannes, 2011; Yaniv, 2004; Yaniv & Milyavsky, 2007), forecasts that were updated after viewing advice became more accurate on average.

The preference patterns for different sources of advice, and the clear benefits to using advice, raise an important question: how can judges be encouraged to use advice? One possibility may be to give judges more autonomy in the process of receiving advice. Past research found that when judges explicate their preferences, it can impact their behavior to align with these explicit preferences more closely (Wilson & Schooler, 1991). Perhaps by allowing judges to opt into advice, and select which source they would prefer to receive advice from, they would be more likely to make use of advice (e.g., Kaufamann & Budescu, 2020; Rabinovitch, et al. 2022). Given the consistent preference for hybrid advice, this raises an intriguing possibility for how hybrid advice might encourage judges to optimize their judgments.

4.2 Similarity Between Initial Beliefs and Advice

Our results provided some novel insights we did not anticipate. In both studies, judges updated their beliefs roughly half the time, while sticking with their initial judgments in the other cases. There was strong evidence that this was done judiciously and strategically, such that judges used the advice mostly when they recognized they were relatively uninformed or naïve. Initial judgments that were relatively accurate prior to viewing advice were less likely to be updated after advice was made available than those that were less accurate.

To probe this effect, we modeled judges' decision to update their beliefs as a function of three classes of variables: personal or demographic variables, features of the items and the situation, and variables representing correspondence between a judge's initial belief and the advice they viewed. A remarkable aspect of our model comparison is that it allowed us to pinpoint the timing of the decision to use advice. We found that the decision is driven primarily by the third set of variables, that are available only after the judges are exposed to the advice.

There were three key predictors of the judges' decisions to revise their beliefs. The first was the distance between a judge's initial forecast distribution and the advice, which had a curvilinear effect on their willingness to update. Judges were less likely to update if the advice was either extremely similar or extremely different from their initial beliefs, but most willing to make use of it for moderate distances. This result is consistent with past research that has found a non-linear effect of the distance between a judges' initial belief and available advice (Allahverdyan & Galstyan, 2014; Schultze et al., 2015; Whittaker, 1963). The second was rank order (dis)agreement between each of the options. Judges were most willing to rely on the advice when it presented counterfactual information, and indicated that the most and least likely options were different from what they initially believed. The third was the relative peakedness of the probability distributions of a judges' initial forecasts and the advice, which we interpreted as a proxy for the relative confidence of the judge compared to advice. Judges were generally more likely to update when the advice posed a more peaked distribution than their initial belief, indicating the advisor had a higher degree of confidence that they did, consistent with findings about the importance of advisor confidence in advice taking (Budescu & Yu, 2007; Van Swol & Sniezek, 2005).

4.3 Accepting Advice

This work suggests a new way of thinking about the judges' roles in a JAS. Research often treats the judge as an agent who, upon receiving persuasive information, decides how to combine their opinions with that advice, and suggests that judges often underutilize advice due to biases such as egocentric trimming, anchoring, and confirmation bias (Allahverdyan & Galstyan, 2014; Bonaccio & Dalal, 2006; Schultze et al., 2015; Yaniv, 2004). Our results suggest that the psychological process underlying this process may consist of two separate stages: first judges must decide whether to acknowledge the advice at all, and if they choose to consider it, then decide how to aggregate their beliefs with the new information. There was even a small minority of participants who fully adopted the advice in a plurality of cases, as opposed to compromise. In Study 1 there were 15 (6%) such participants, and in Study 2 there were three (2%).

Our results imply several strategies for dissemination of advice such that judges may be most inclined to maximize its use. Judges appear most amenable to accepting advice when (a) it is moderately different from their own beliefs, (b) it provides counterfactual information, and (c) the advisor is more confident than they are. When any of these conditions is unmet, there may be little, if any, benefit to providing judges with advice. It may be worth studying these precise conditions in a more controlled experimental fashion as well. Schultze et al. (2015) have already conducted such a study for distance between initial belief and advice.

Another implication of our results is that judges are quite astute in recognizing cases where they need, and are likely to benefit from, advice. It may be beneficial to set up JAS in which advice is optional and is presented only when the judge considers it necessary and ask for it. Such a setup would be more efficient (our data indicate that it would cut the number of times advice is offered in half!), and it may also increase the impact of the advice because the judges would be more likely to follow it. Kaufmann and Budescu (2020) present some preliminary support for this hypothesis, but it should be studied carefully in additional settings.

A defining feature of our (and most JAS) studies was that judges were required to make their beliefs explicit prior to receiving advice. This is a convenient way to study how beliefs change in light of advice, but it is likely that having judges explicate their belief affects how they incorporate new information, and some differences have been observed in studies in which pre-advice beliefs were elicited as compared to studies in which they were not (Bonaccio & Dalal, 2006; Rader et al., 2015; Sniezek & Buckley, 1995). It may be worth considering alternative designs in which judges' beliefs are not made explicit.

4.4 Methodological Contributions

Most of the past research on advice utilization involved scenarios where judges estimated a single value or a point estimate of a continuous variable. Our research involves forecasts where, in many cases, judges estimate a multinomial probability distribution. This required devising a generalization of the well-established WOA metric to a version based on distances in multidimensional space, which we term DWOA. Advice utilization for these multinomial probability judgments did not differ qualitatively from patterns in past literature, or patterns for items with only two bins. However, it may be possible to define alternative, or even superior generalizations of WOA that satisfy the same key properties as DWOA, and a more dedicated methodological investigation into the problem would be a worthwhile task for future research.

4.5 Limitations

Despite its many novel contributions, this work has several limitations. Most notably, the types of geopolitical forecasting problems studied are highly idiosyncratic. It is unrealistic to expect the volunteer participants had strong *a priori* beliefs about most of the items they forecasted. Reference links were provided to allow participants to do preliminary research on each topic, but it is difficult to know if these results might generalize to topics with which participants might have more direct familiarity, or less difficult topics to forecast (Gino & Moore, 2007). The definition of domain was also admittedly somewhat subjective, as well as potentially confounded by the quantitative nature of many of the economic items, and more qualitative nature of the politics items. Future research could attempt to more directly tease out the effects of these confounds. And as noted previously, although the differences in time horizon were more objective than domain, study of even longer time horizons remains an important area of research in forecasting (see Himmelstein et al., In Press), and may explain why certain effects reported in prior research were not apparent in our results. Finally, a more principled methodological investigation of multinomial generalizations to belief revision measurement could provide a stronger foundation for our empirical results.

5 Conclusion

Our study was motivated by the debate on algorithm appreciation or aversion. We found that in the context of forecasting geopolitical events with relatively short time horizons, there is no evidence for aversion to either source. The judges' belief revisions in our studies were driven by the quantitative information conveyed by the advice, and its similarity to their prior beliefs, rather than the source that generated the advice. These results provide an important addition not only to the literature on algorithmic scaffolding, but also to the literature on social persuasion. We found clear evidence of differential attitudes towards algorithmic advice for different content domains, however these attitudes did not map into judges' willingness to use the advice when making incentivized judgments, presumably because both sources are perceived sufficiently trustworthy. Several clear patterns emerged involving the similarity between the advice and people's initial beliefs. These patterns involved not only the weight people put on the advice, but also simply their decision to make use of it in general. These results provide novel insights and imply several new directions for future research.

6 Data Disclosure

All sample sizes were determined based upon available funding, such that participants would be paid at a rate of at least $15 per hour based on the expected time to survey completion conveyed in the informed consent document. Both because these constraints allowed us to collect relatively large samples, and because the key analyses were based on hierarchically defined models, for which power *a priori* power analysis is not straightforward without a principled *a priori* estimate of intra-class correlation (Hedges & Hedberg, 2007), power analysis was not conducted prior to data collection. Upon completion of Experiment 1, we conducted two power analyses to determine the minimum effect sizes that we would have 0.80 and 0.90 power to detect with $\alpha = .05$, based on the Satterthwaite adjusted degrees of freedom (rounded down to the nearest 100 in cases where adjusted DF > 100).

The first power analysis was based on the three-way ANCOVA for advice source, time horizon, and domain on belief revision in the JAS task, with a focus on interactions and main effects involving advice source. Results of the power analysis indicated we had 0.80 power to detect $\eta^2 = .002$[16] for the three-way interaction between advice source, domain, and time horizon; $\eta^2 = .002$ for the two-way interaction between advice source and domain; and $\eta^2 = .002$ for the main effect of advice source. With 0.95 power, the corresponding values were $\eta^2 = .003$, $\eta^2 = .003$, and $\eta^2 = .004$. Each of these detectable effect sizes is extremely small, thus we conclude the study was sufficiently powerful.

The second power analysis was for the two-way ANCOVA in the preferential task. Because the dependent variable was a bi-polar scale anchored by preference for human or algorithmic advice, we conducted power analysis on the two-way interaction and main effects of time horizon and domain. Results of the power analysis indicated we had .80 power to detect $\eta^2 = .004$ for the two-way interaction between time horizon and domain; $\eta^2 = .004$ for the main effect of time horizon; and $\eta^2 = .60$ for the main effect of domain. With .95 power, the corresponding values were $\eta^2 = .007$, $\eta^2 = .006$, and $\eta^2 = .71$. Because the detectable effect

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[16] Reported effect sizes are in ω^2, which is a more conservative adjustment to η^2

sizes involving time horizon were extremely small, and the only significant effect was the main effect of domain ($\omega^2 = .82$, p < .001) we conclude the study was sufficiently powerful. Results of both power analyses are also supported by successful replications in Study 2.

The data reported in the manuscript and supplementary materials represent all data collected with the following exceptions:

In Study 2, a separate portion of the study was conducted as an exploratory investigation into the validity of a proposed Coherence Forecasting Scale (CFS). Because this was a separate research project and in the exploratory validation stage, these results are not reported (see Ho, 2020).

The original plan for Study 2 was to use Qualtrics Panel participants, as they offered us reduced fees (beyond participant compensation) compared to MTurk. However, preliminary analysis of the CFS, CRT, and SNS scales, as well as the preference ratings revealed a large proportion of straight-lined results and extremely short response times. This led us to conclude the data quality was too poor to be useful. None of the actual forecast data was subsequently analyzed (e.g., DWOA or Brier Scores were never calculated), these data were discarded, and data collection switched back to the MTurk platform.

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Tables

Table 1

Accuracy analysis in cases where judges updated their original forecasts in Study 1

	No Update			Update									No Update – T2
				T1			T2		T1 - T2				
Subset	N	Mean Brier	SE	N	Mean Brier	SE	Mean Brier	SE	Mean Improvement	SE	D	Proportion Improved	D
Short	1227	0.353	0.009	1149	0.468	0.011	0.337	0.007	0.131	0.008	0.483	0.672	0.046
Long	1166	0.401	0.010	1208	0.483	0.010	0.383	0.007	0.100	0.008	0.360	0.618	0.052
Econ	944	0.414	0.010	1056	0.465	0.010	0.374	0.007	0.090	0.007	0.396	0.693	0.126
Politics	1085	0.332	0.011	915	0.49	0.013	0.330	0.01	0.160	0.010	0.529	0.658	0.005
Human	1218	0.377	0.010	1158	0.474	0.011	0.359	0.008	0.115	0.008	0.422	0.655	0.050
Machine	1175	0.376	0.010	1199	0.477	0.010	0.362	0.007	0.115	0.008	0.415	0.633	0.041
Overall	2393	0.376	0.007	2357	0.476	0.007	0.361	0.005	0.115	0.006	0.395	0.644	0.044

Table 2

Likelihood Ratio Tests for Mixed Effects Logistic Regressions in Study 1

		Model 1			Model 2			Model 3		
Variable Class	Effect	χ^2	df	p	χ^2	df	p	χ^2	df	p
Individual Differences	Age	1.99	1	.16	1.61	1	.20	0.87	1	.35
	Gender	0.03	1	.85	0.03	1	.87	0.00	1	.97
	Stats Experience	0.82	1	.37	1.01	1	.32	1.14	1	.29
Experimental Manipulations	Advice Source				2.18	1	.14	1.57	1	.21
	Domain				5.59	2	.06	8.13	2	.02
	Time Horizon				2.95	1	.09	4.25	1	.04
	Advice Source*Domain				0.26	2	.88	0.64	2	.73
	Advice Source*Time Horizon				3.27	1	.07	2.32	1	.13
	Domain*Time Horizon				0.31	2	.86	0.16	2	.92
	Advice Source*Time Horizon*Domain				1.18	2	.55	0.42	2	.81
T1 and Advice Relationship	$D(F_1, A)$							172.07	1	<.001
	$D(F_1, A)^2$							46.11	1	<.001
	Discrimination Agreement							37.01	1	<.001
	Relative Sharpness							41.54	1	<.001
	$D(F_1, A)$*Agreement							26.72	1	<.001
	$D(F_1, A)^2$*Agreement							33.95	1	<.001
	$D(F_1, A)$*Sharpness							8.18	1	.004
	$D(F_1, A)^2$*Sharpness							8.98	1	.003
	Agreement*Sharpness							2.72	1	.10
Model Fit	BIC		5290.53			5366.69			4562.52	
	AUC (fixed effects only)		0.53			0.57			0.66	
	AUC (fixed and random effects)		0.86			0.86			0.92	

Note: Significant effects (p < .01) are highlighted

Table 3

Accuracy analysis in cases where judges did or did not update in Study 2

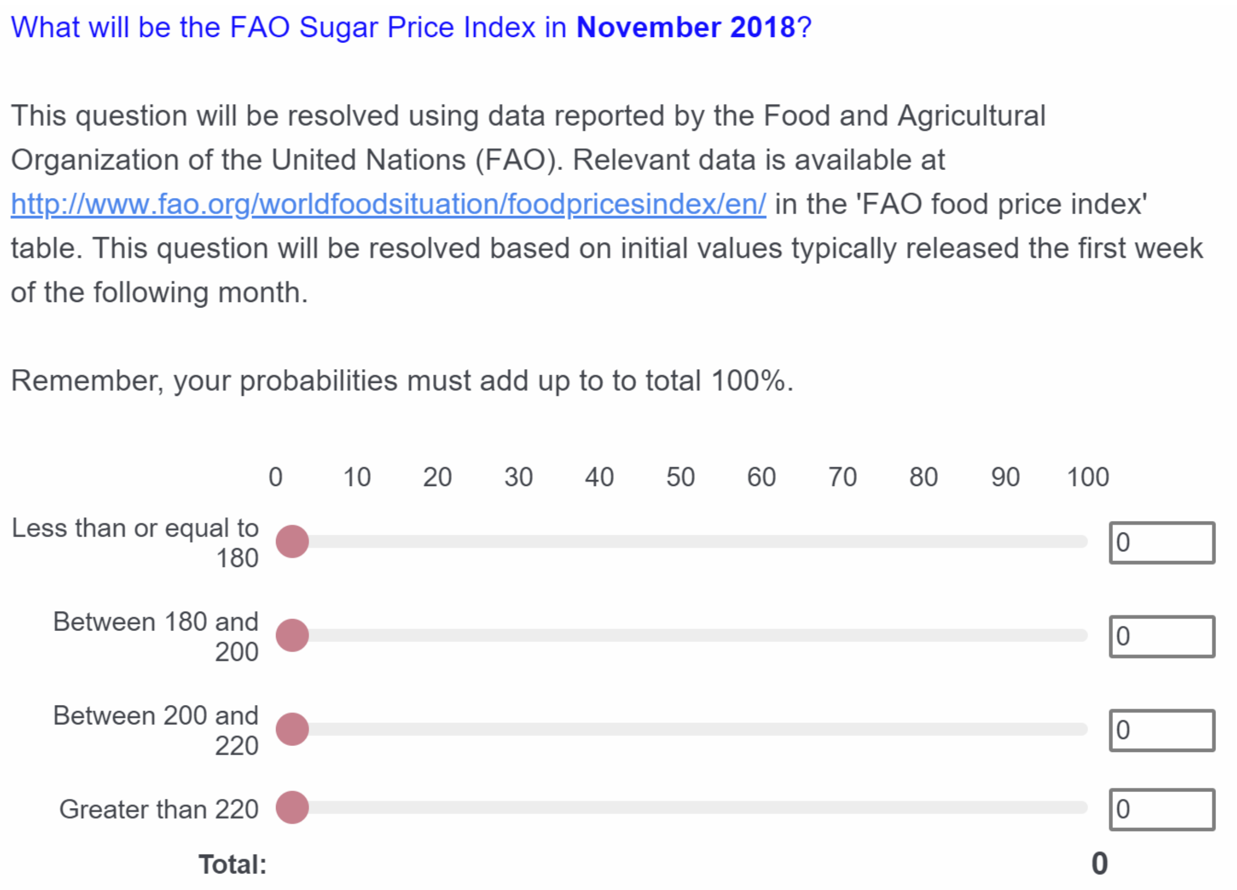
	No Update			Update									No Update – T2
				T1			T2		T1 - T2				
Subset	N	Mean Brier	SE	N	Mean Brier	SE	Mean Brier	SE	Mean Improvement	SE	D	Proportion Improved	D
Econ	504	0.481	0.015	522	0.540	0.014	0.487	0.009	0.052	0.010	0.232	0.534	-0.022
Politics	557	0.231	0.010	469	0.366	0.015	0.242	0.007	0.124	0.014	0.375	0.563	-0.056
Human	426	0.360	0.016	429	0.440	0.015	0.366	0.011	0.074	0.012	0.299	0.536	-0.022
Machine	435	0.362	0.015	420	0.458	0.018	0.366	0.011	0.092	0.015	0.294	0.531	-0.015
Hybrid	449	0.363	0.016	406	0.438	0.016	0.363	0.010	0.075	0.013	0.272	0.544	0.000
Overall	1310	0.362	0.009	1255	0.445	0.010	0.365	0.006	0.080	0.008	0.276	0.531	-0.011

Table 4

Likelihood Ratio Tests in Study 2

		Model 1			Model 2			Model 3		
Variable Class	Effect	χ^2	df	p	χ^2	df	p	χ^2	df	p
Individual Differences	Age	0.01	1	.92	0.01	1	.92	0.03	1	.86
	Gender	1.80	1	.18	1.81	1	.18	2.34	1	.13
	Stats Experience	<0.00	1	>.99	<0.00	1	>.99	0.04	1	.84
	CRT	3.36	1	.07	3.36	1	.07	5.15	1	.02
	SNS	0.18	1	.67	0.18	1	.67	0.09	1	.77
Experimental Manipulations	Advice Source				1.69	2	.43	0.92	2	.63
	Domain				1.79	2	.41	6.04	2	.05
	Advice Source*Domain				2.31	4	.68	3.23	4	.52
T1 and Advice Relationship	$D(F_1, A)$							136.94	1	<.001
	$D(F_1, A)^2$							43.02	1	<.001
	Discrimination Agreement							11.74	1	<.001
	Relative Sharpness							54.15	1	<.001
	$D(F_1, A)$*Agreement							2.19	1	.14
	$D(F_1, A)^2$*Agreement							3.53	1	.06
	$D(F_1, A)$*Sharpness							9.33	1	.002
	$D(F_1, A)^2$*Sharpness							9.27	1	.003
	Agreement*Sharpness							22.13	1	<.001
Model Fit	BIC	2834.41			2891.27			2539.55		
	AUC (fixed effects only)	0.53			0.57			0.66		
	AUC (fixed and random effects)	0.88			0.88			0.93		

Figures



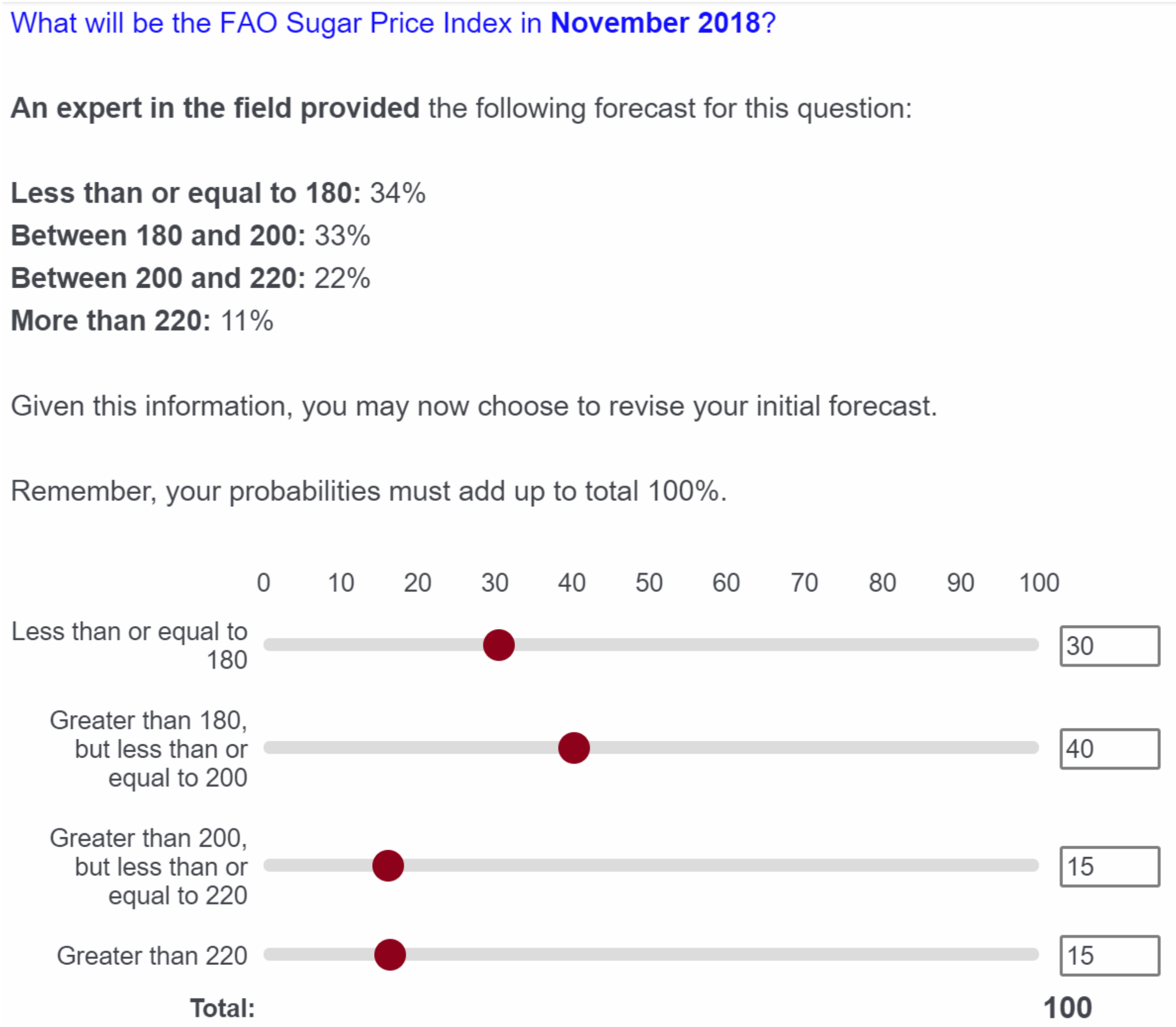
What will be the FAO Sugar Price Index in **November 2018**?

This question will be resolved using data reported by the Food and Agricultural Organization of the United Nations (FAO). Relevant data is available at http://www.fao.org/worldfoodsituation/foodpricesindex/en/ in the 'FAO food price index' table. This question will be resolved based on initial values typically released the first week of the following month.

Remember, your probabilities must add up to to total 100%.

	0 10 20 30 40 50 60 70 80 90 100	
Less than or equal to 180		0
Between 180 and 200		0
Between 200 and 220		0
Greater than 220		0
Total:		**0**

Figure 1 – Example of forecasting question from Pilot Study and original judgement in Study 1.



What will be the FAO Sugar Price Index in **November 2018**?

An expert in the field provided the following forecast for this question:

Less than or equal to 180: 34%
Between 180 and 200: 33%
Between 200 and 220: 22%
More than 220: 11%

Given this information, you may now choose to revise your initial forecast.

Remember, your probabilities must add up to total 100%.

	0 10 20 30 40 50 60 70 80 90 100	
Less than or equal to 180		30
Greater than 180, but less than or equal to 200		40
Greater than 200, but less than or equal to 220		15
Greater than 220		15
Total:		**100**

Figure 2. Advisory forecast and revision elicitation from the second stage of Study 1.

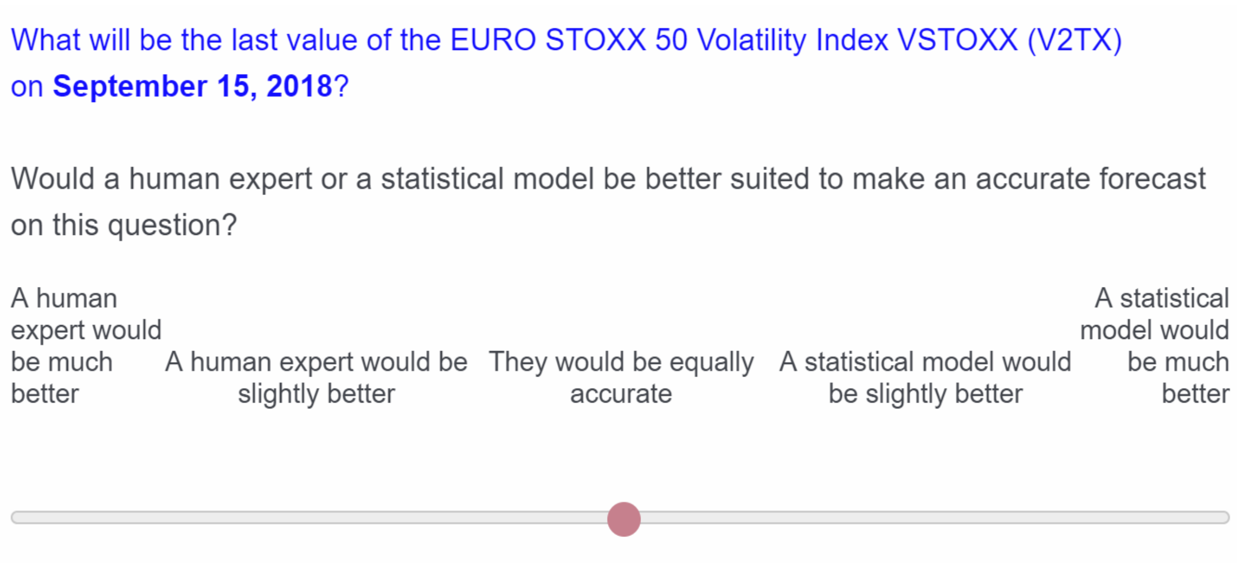


Figure 3. Example of preference task item from Study 1

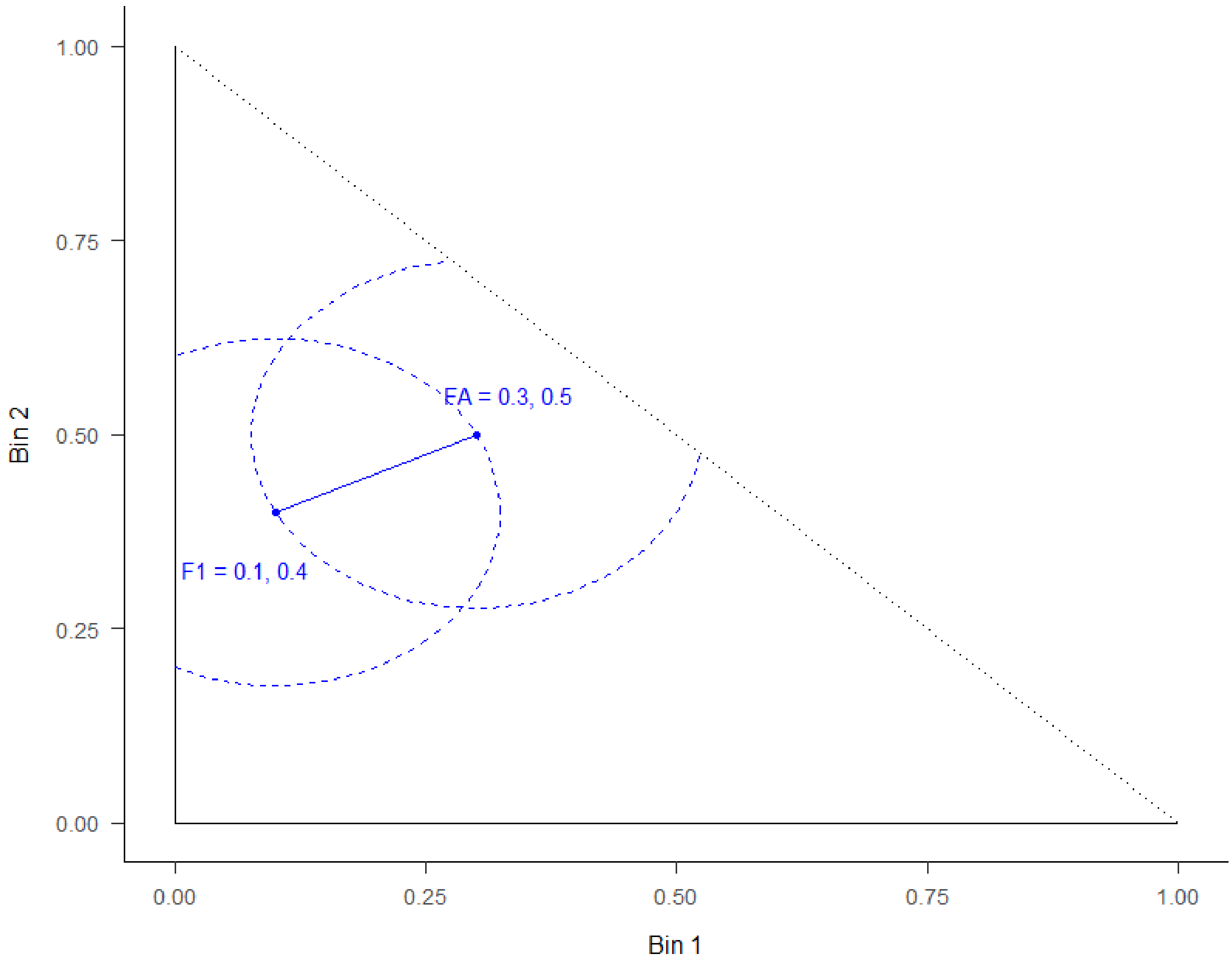


Figure 4. Graphical representation of DWOA in a two-dimensional simplex. In this example $F_1 = (.1, .4, [.5])$ and $F_a = (.3, .5, [.2])$. The line connecting the two points represents D(F1, FA). The region where the two circles overlap represents all possible values of F_2 such that D(F2, FA) ≤ D(F1, FA) and where D(F1, F2) ≤ D(F1, FA). Results outside of this region, which represent approximately 9% of all observations in both Study 1 and Study 2, are excluded from the main analysis. Analyses with these observations included can be found in Appendix XX

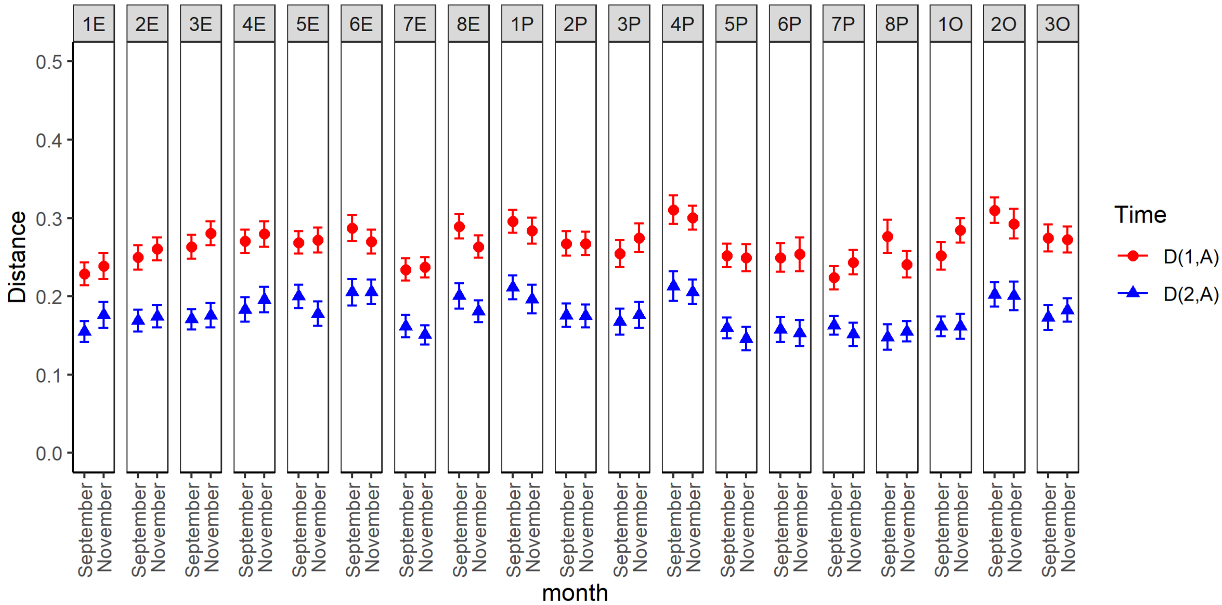


Figure 5. Mean distance between advice and the first forecast, $D(F_1, F_a)$, and between advice and the second forecast, $D(F_2, F_a)$, for each item with standard errors. Note: Item labels: E = Economics, P = Politics, O = Other

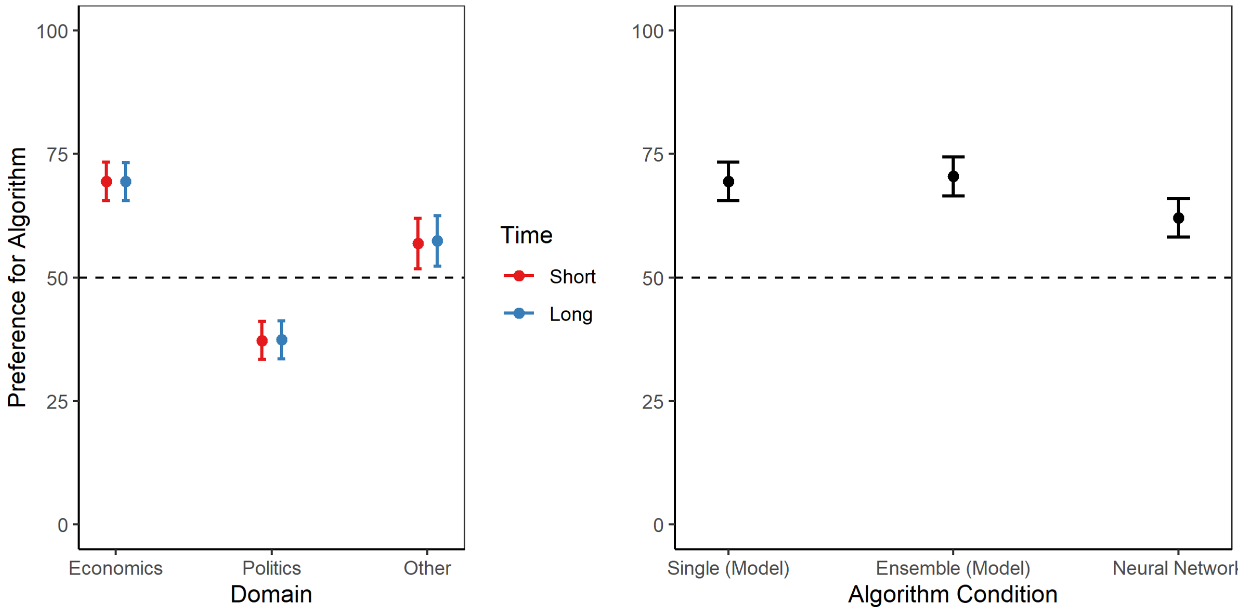


Figure 6. Left Panel*:* Mean preference for algorithmic advice as a function of domain and time horizon in Study 1. Right panel: Mean preference for algorithmic advice under three descriptions of the algorithm. The horizontal line at 50 indicates no preference for either source.

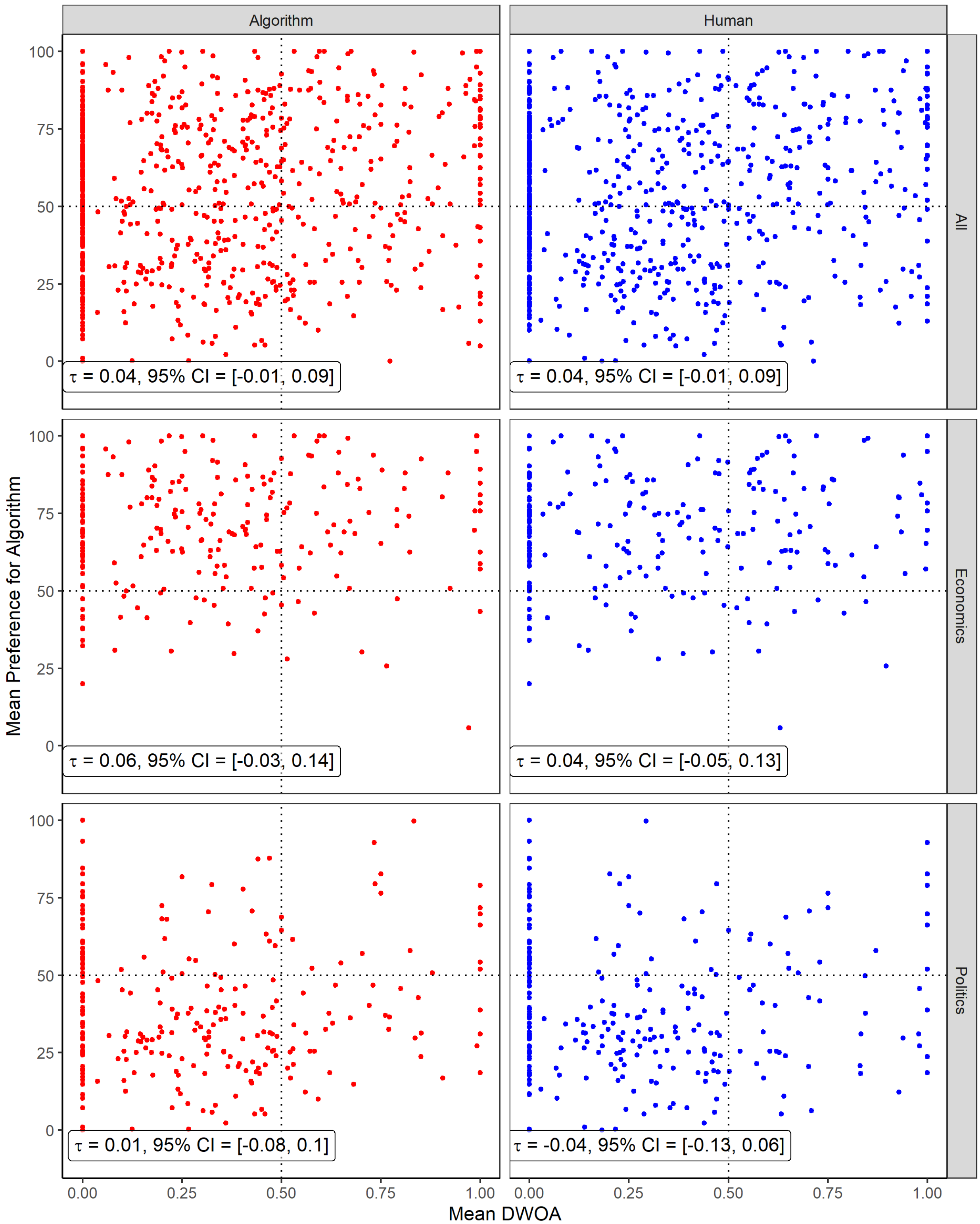


Figure 7. Joint distributions of DWOA and preference ratings in Study 1 with Kendall's Tau rank correlation and 95% bootstrapped confidence intervals.

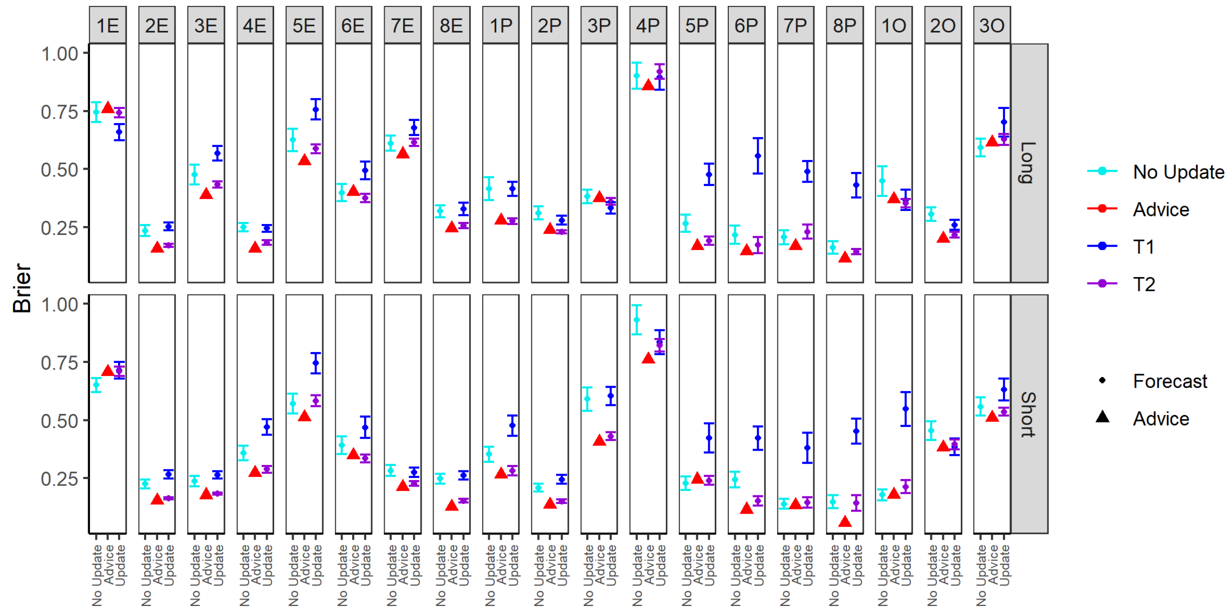


Figure 8. Brier scores (lower scores are better) for all items as a function of time of forecasts and taking advice in Study 1

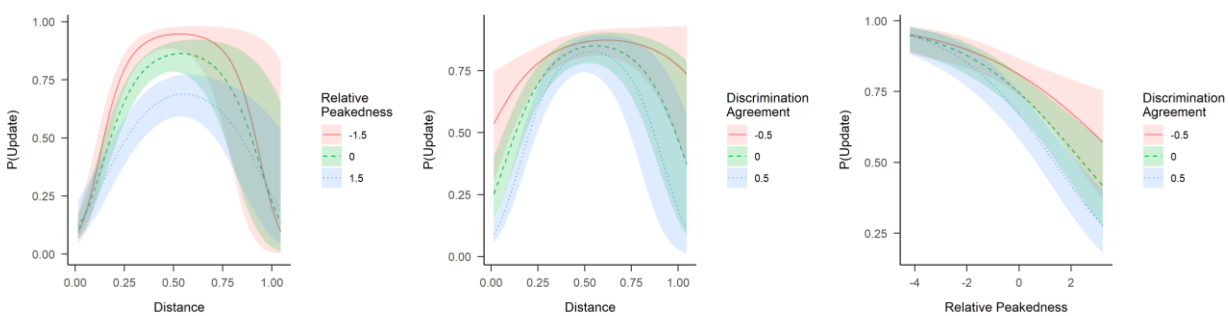


Figure 9. Left Panel: Probability to update forecasts as a function of $D(F_1, F_a)$ and relative peakedness in Study 1. Center Panel: Interaction between discrimination and distance in Study 1. Right Panel: Interaction between discrimination and relative peakedness in Study1.

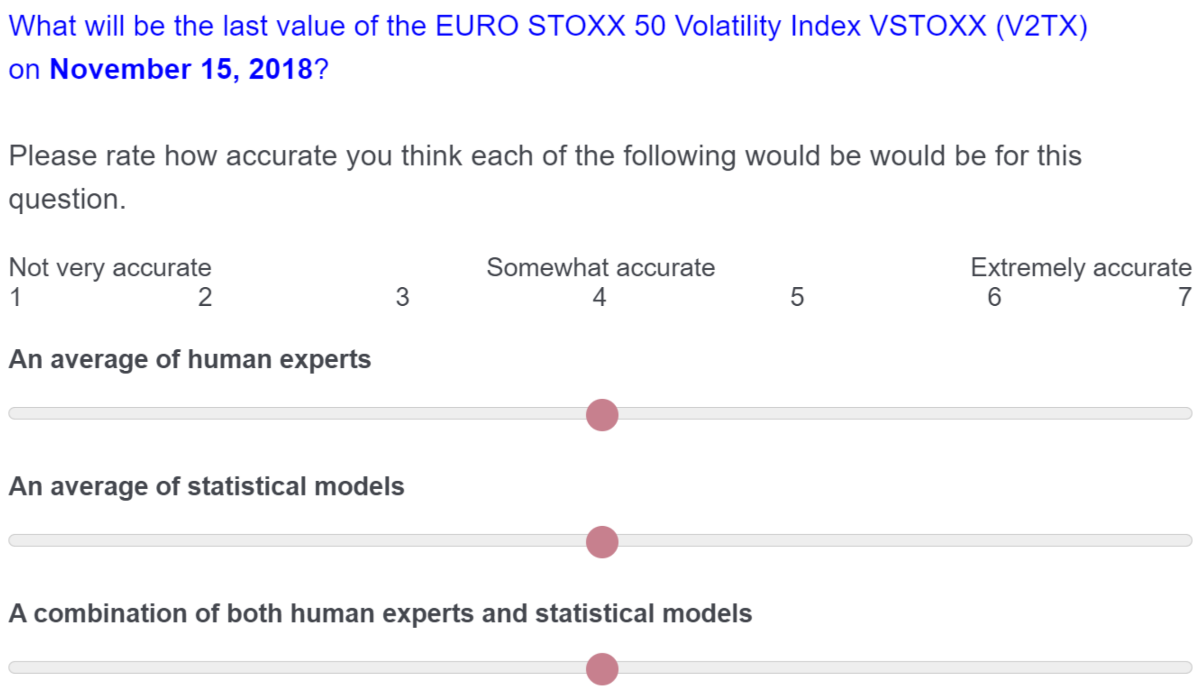


Figure 10 Preferential task in Study 2

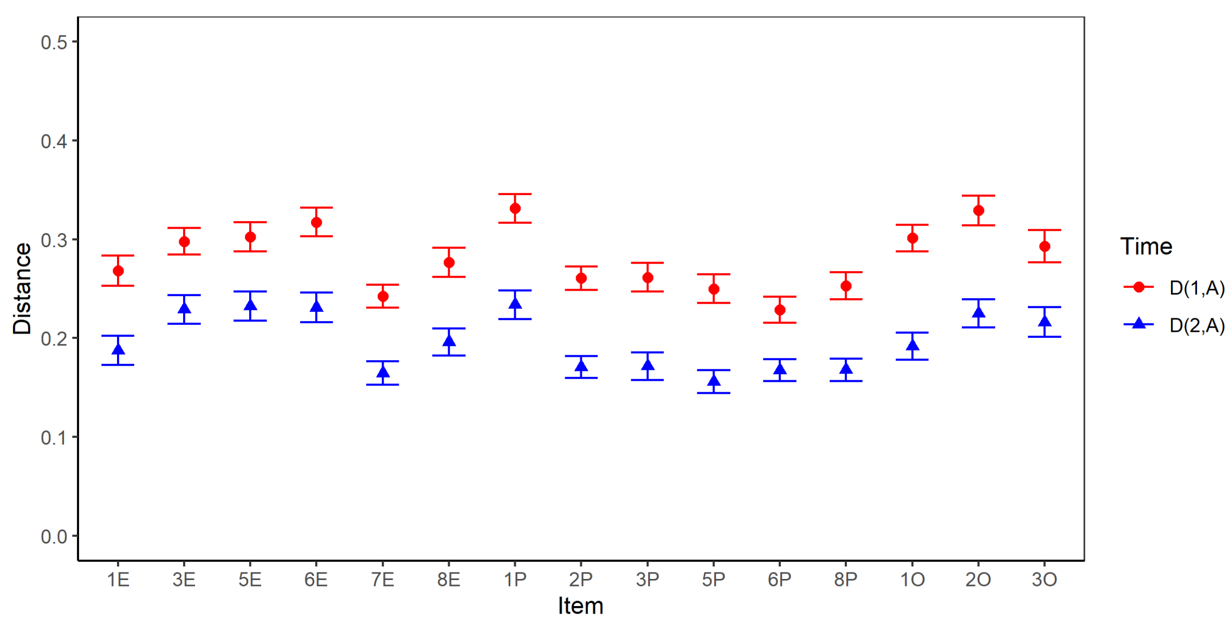


Figure 11. Mean distance between advice and the first forecast, $\mathrm{D}(F_1, F_a)$, and between advice and the second forecast, $\mathrm{D}(F_2, F_a)$, for each item with standard errors in Study 2.
Note: Item labels: E = Economics, P = Politics, O = Other

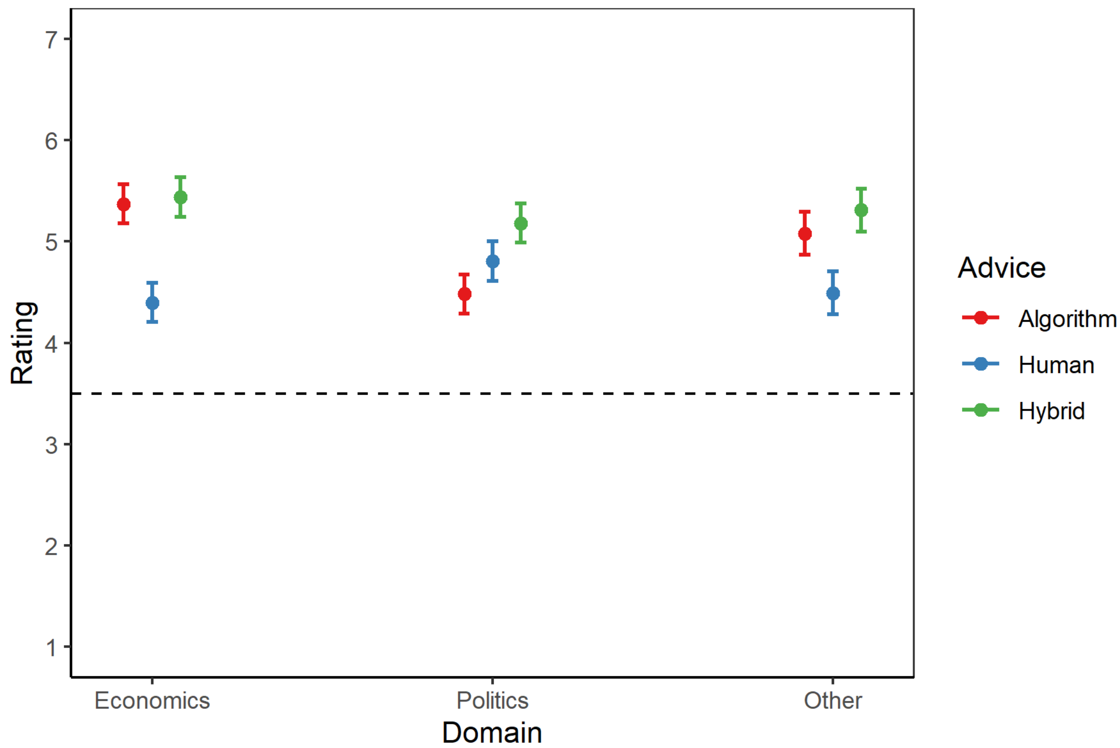


Figure 12 Mean ratings of the three sources and the various domains in Study 2

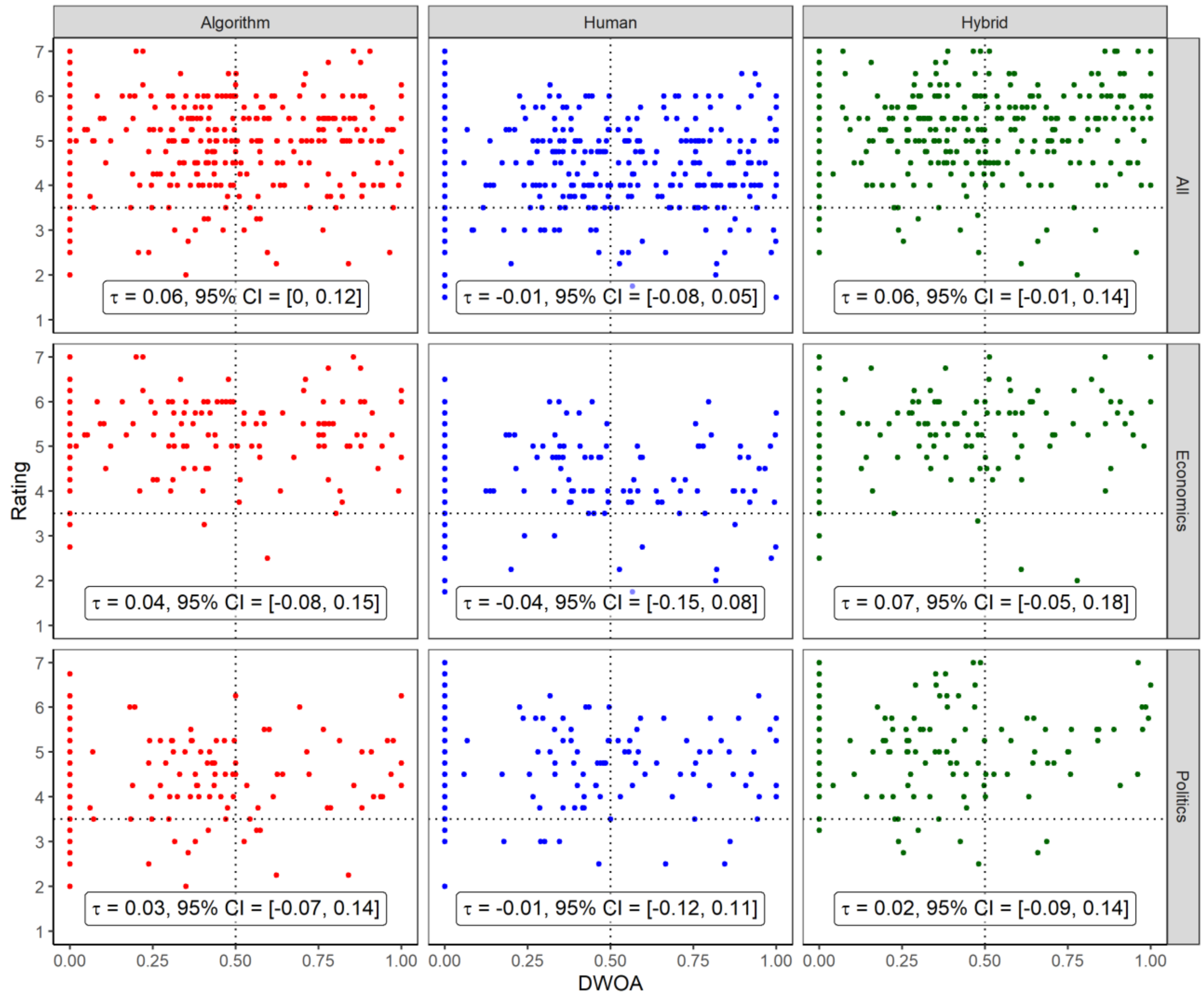


Figure 13. Joint distributions of DWOA and preference ratings in Study 2 with Kendall's Tau rank correlation and 95% bootstrapped confidence intervals.

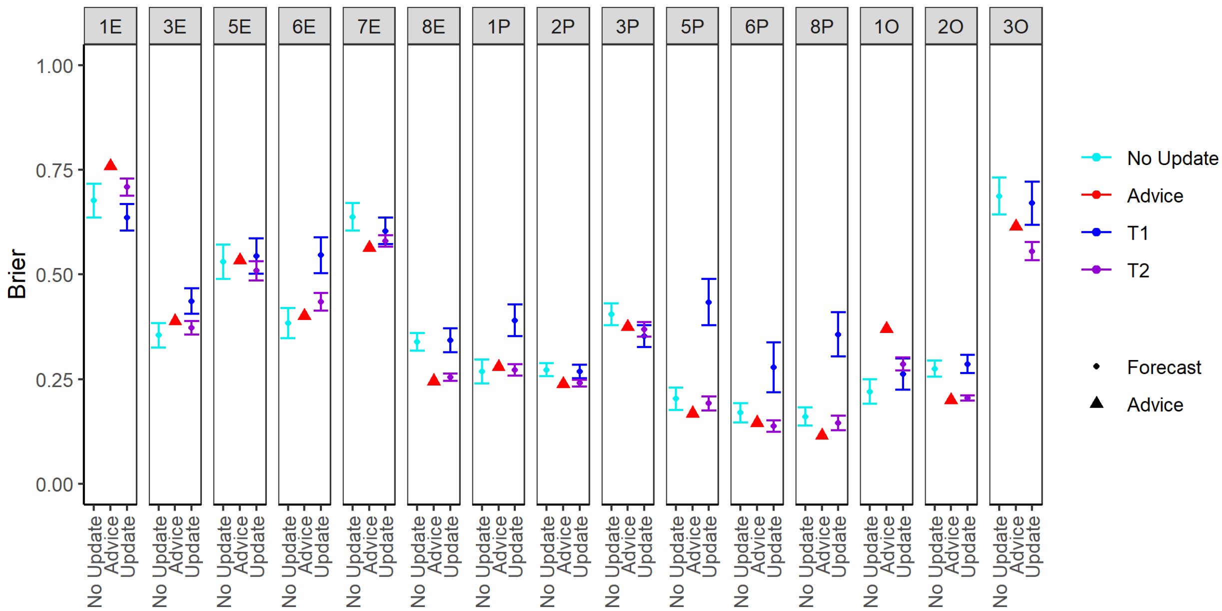


Figure 14. Brier scores (lower scores are better) for all items as a function of time of forecasts and taking advice in Study 2

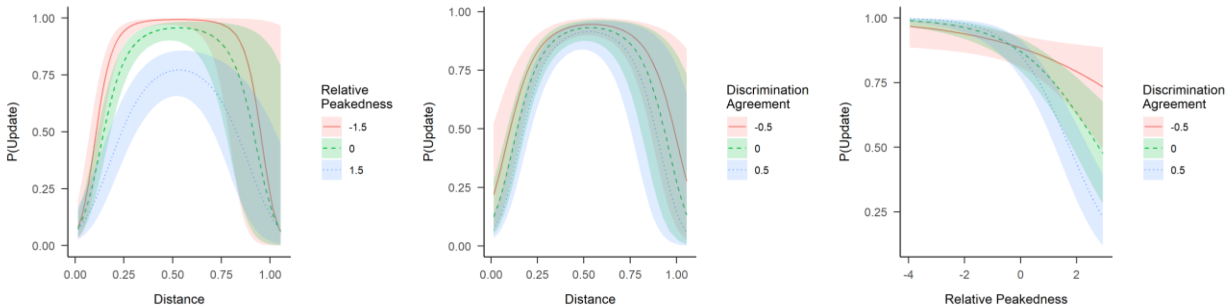


Figure 15. Left Panel: Probability to update forecasts as a function of $D(F_1, F_a)$ and relative peakedness in Study 2. Center Panel: Interaction between discrimination and distance in Study 2. Right Panel: Interaction between discrimination and relative peakedness in Study 2.